

/ UNEXPECTED CONNECTIONS /

How a vulnerability in obscure
dealer software could have
unlocked your car from anywhere



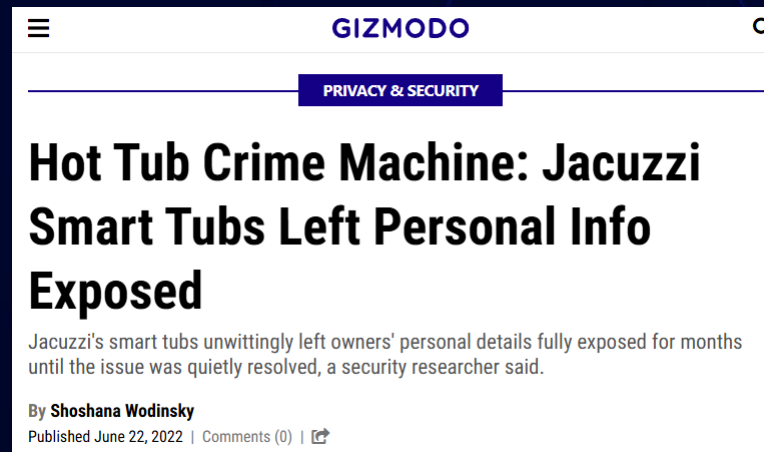
EATONWORKS



harness

/ me

- A constant thorn in automakers' side (:
- Started with game hacking (Game Shark, Action Replay, Xbox modchips)
- First big automaker hack (Toyota supplier network) blew up in the news



McDelivery

/ Automaker Motivation

- Just for fun!
- Started with Toyota (we own Toyota cars)
- Lots of domains = big attack surface

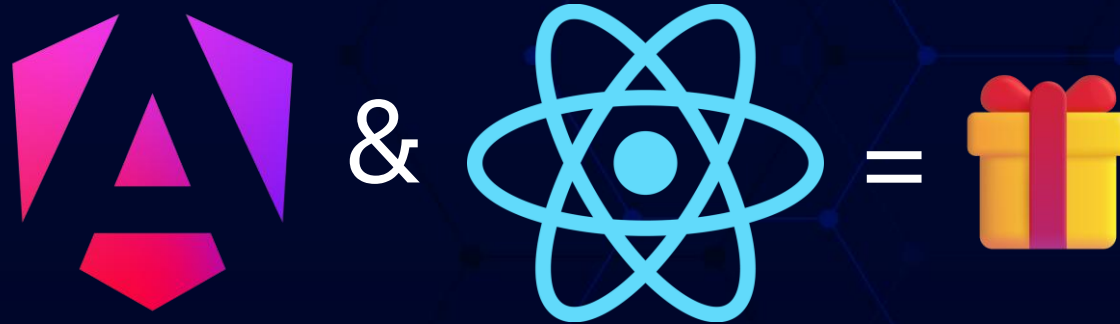
Amazon	d565e29ae3e248	*.mbbrandhub-nonprod.nprd.eucl.sys.cbenz.com	2023-07-25	2024-08-22	non-EV
Amazon	9f42b9854c2e79f	*.mbbrandhub-prod.prd.eucl.caas.onewebenz.com	2023-07-25	2024-08-22	non-EV
Amazon	8921f789d9a885C	*.retail.aem.oneweb.merbenz.com	2023-07-31	2024-08-28	non-EV
Amazon	5f11a04e15e3c8ed	*.vpcu-nonprod.nprd.eucl.sys.cbenz.com	2023-08-03	2024-08-31	non-EV
Amazon	2109554a35aeb9	*.vpcu-nonprod.nprd.eucl.caas.benz.com	2023-08-03	2024-08-31	non-EV
Amazon	9a87215f1a282e2f	*.monitoring-nonprod.nprd.eucl.caas.benz.com	2023-08-06	2024-09-04	non-EV
Amazon	6a81b568e937ec2	*.monitoring-nonprod.nprd.eucl.sys.cbenz.com	2023-08-06	2024-09-04	non-EV
Amazon	7ed453db451054	*.tools-nonprod.nprd.eucl.sys.cbenz.com	2023-08-07	2024-09-05	non-EV

Page 1 of 42 100 items per page 1 - 100 of 4188 items

CT logs from an automaker showing thousands of subdomains.

/ What I look for

- JS heavy apps (especially with source maps)
- Ripe for API and secret discovery
- Admin functionality can also be uncovered
 - Remember this point!



Angular & React apps are great hack targets!

/ What I hacked

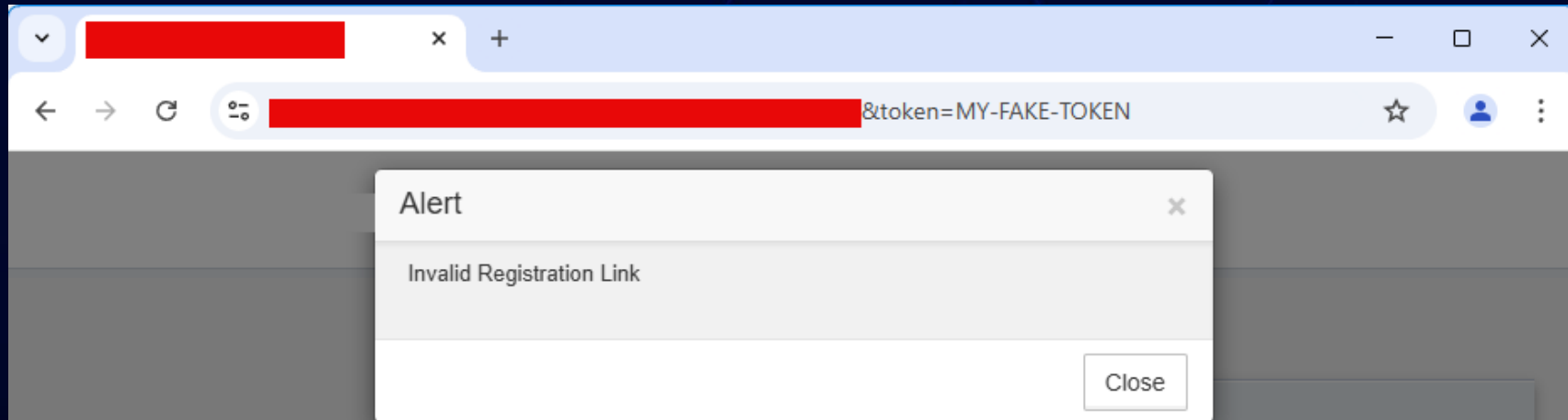
- ✨ Brand new, DEF CON 33-exclusive vulnerability disclosure! ✨
- Found vulnerabilities in top automaker's centralized dealer platform
- Used by automaker's authorized dealers in the USA
- End result: creation of national admin account
- Full access to **over 1,000** USA dealers
- Could start your car, get your personal info, & more

/ Dealer System Details

- Used to order cars, make sales, manage customers/leads, etc
- Accessible via internet
- Cornerstone of day-to-day operations
- Invite only, no open registration
- Uses two-factor auth
- SAP/Java with AngularJS frontend

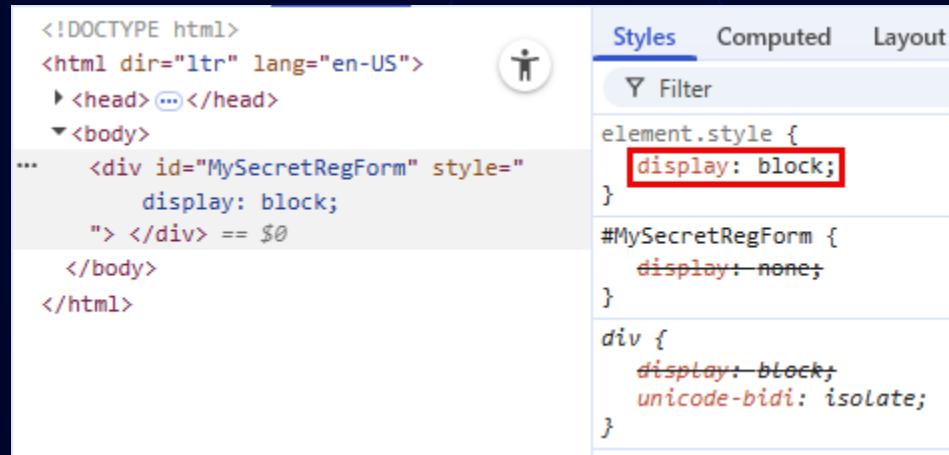
/ The Invite System

- New employee? Dealer admin invites you via email
- Email has unique invite token in link
- Login page has JS to call server-side token validation API
 - Valid token? Registration form shown
 - Bad token? Error!



/ CSS'ing the registration form

- Can't forge token = can't pass API check
- Discovery: hidden registration form in HTML
- Hack: apply “*display: block;*” using Chrome devtools
 - Same end result as passing API check



/ The registration form – unmasked!

Create an Account

Contact Information

* Title

Select

* U.S. Military Veteran

Select

* First Name / MI

/

* Last Name

* Email

EMAIL@EXAMPLE.COM

* Confirm Email

Mobile

Mobile must be 10 digits

* Security Question

Select

* Your Answer

Terms of Use

☐ I Agree To the

Terms of Use

Dealer Access

Select Dealer(S)

Dealer

Verify

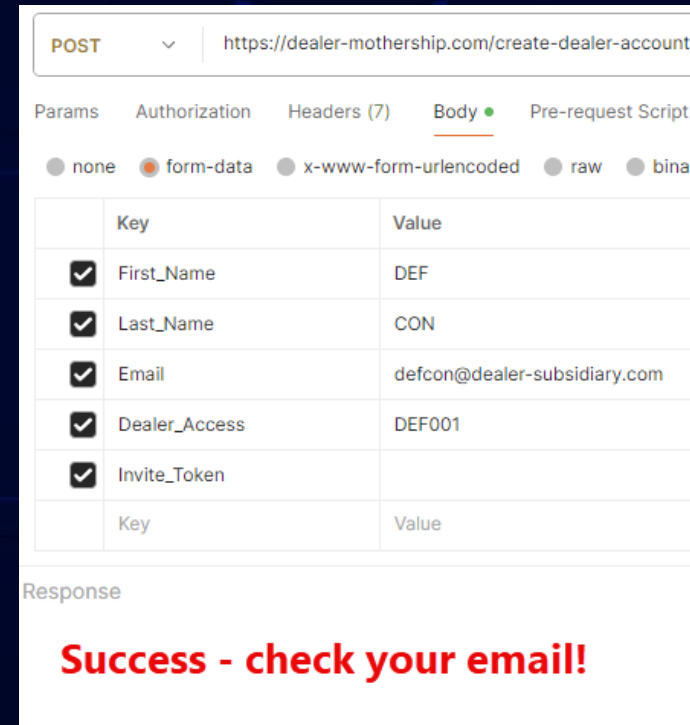
Submit

Cancel

9/77

/ Submitting the registration

- Bad news: Form POST fields include invite token
- Good news: Server doesn't check it!
- Can be blank - anyone could register!



The screenshot shows a Postman interface for a POST request to `https://dealer-mothership.com/create-dealer-account`. The 'Body' tab is selected, and the 'form-data' radio button is chosen. A table lists the form fields with their keys and values. Below the table, the 'Response' section displays the message 'Success - check your email!' in red text.

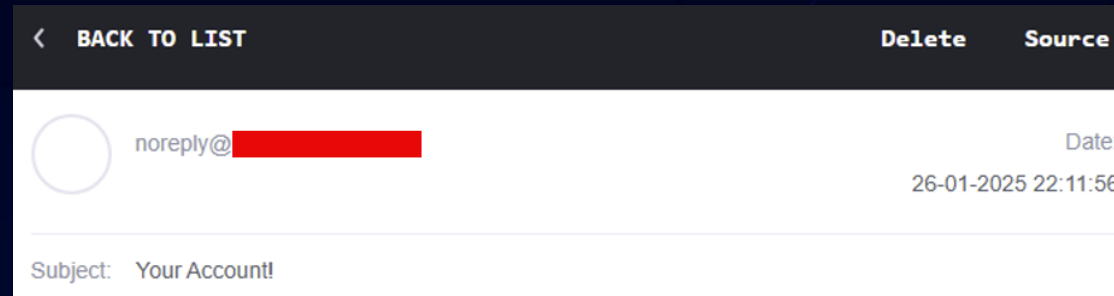
Key	Value
☒ First_Name	DEF
☒ Last_Name	CON
☒ Email	defcon@dealer-subsiary.com
☒ Dealer_Access	DEF001
☒ Invite_Token	

Response

Success - check your email!

Pseudo example in Postman

/ Congratulations! (maybe?)



Congratulations!

You are receiving this email because your profile has been updated. You have completed the registration for a new [redacted]. This ID will allow you to access [redacted] and other related systems.

Please, **DO NOT SHARE** this ID with other people. Use the below URL to access the application and change your temporary password.

Please Note: You will not be able to log in to [redacted] until your Dealer Admin grants you the proper access.

SYSTEM ACCESS INFORMATION

! Authentication failed. Not assigned the requested dealership or Dealer Code incorrect. If this problem persists, contact your dealer administrator.



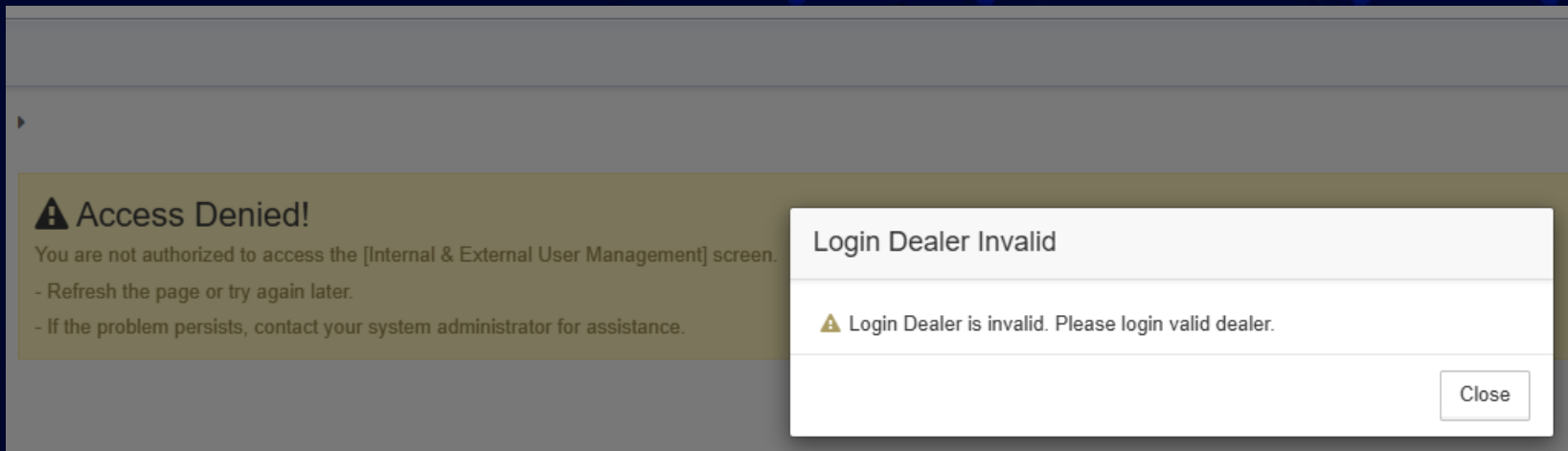


/ What now?

- Account useless - no permissions, no chance of approval
- Almost moved on/gave up
- Login page - limited attack surface
- Rest of site login-protected
- Valid session needed to poke further

/ Profile update breakthrough

- Normal login would not establish a session
- “Profile update” option **did** establish a session! (JSESSIONID)
- Opened the door a tiny bit. Front end now loads!



/ New attack surface

- *Hundreds* of scripts containing APIs/pages ripe for exploit
- Previously unobtainable
- Valid session = hope of exploiting vulnerable APIs

```
'use strict';

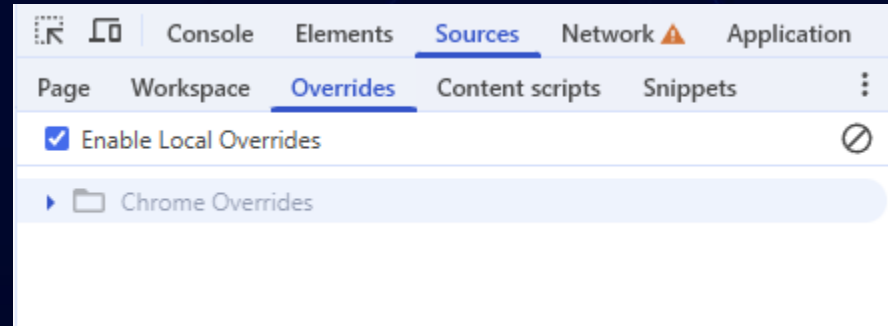
/**
 * Internal & External Information
 */
angular.module('████████.adminl.userManagement', ['████████.common', 'ui.router', 'ui.bootstrap'])
    .config(function($stateProvider) {
        $stateProvider.state('████████.adminl.userManagement', {});

        // Dealer User Inquiry
        stateProvider.state('████████.adminl.userManagement.inExUserSearch', {
            url: '/adminl_inExUserSearch',
            data: {
                breadcrumbs: ['Admin()', 'User Management', 'Internal & External User Management'],
                title: 'Internal & External User Inquiry'
            }
        });
    });
```

Discovered admin script

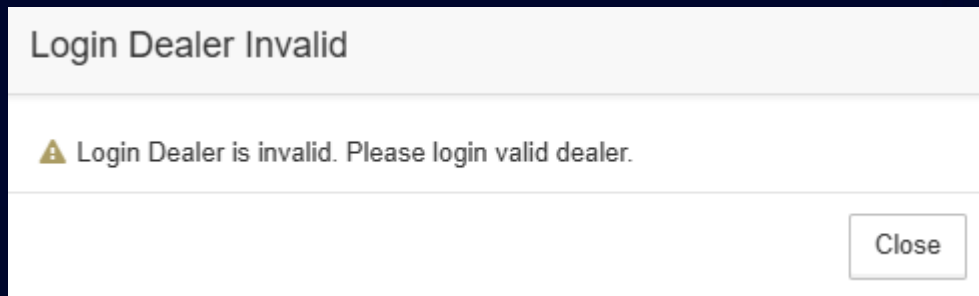
/ Patching JS

- My favorite Chrome feature: Local Overrides
- Lets you override scripts from websites
- You can modify/patch things, like client-side security checks



/ Accessing admin user management

- Can't access normally; necessary to patch scripts to bypass errors
- First patch: comment out annoying popup



```
// Login Dealer Code Valid
/*if (retData.loginDealerCodeValid == false) {
    commonUtil.alertModalWarning({
        title: $rootScope.██████████Lang.common.tx_loginDealerInvalid,
        content: $rootScope.██████████Lang.common.tx_loginValidDealer,
        showClose: false
    });

    // Logout
    setTimeout(function(){
        commonUtil.clearToken();
        $('#logoff').submit();
    }, 5000);
}*/
```


/ Accessing admin user management

- Second patch: bypass access denied display
- Returning empty string means you have access

⚠ Access Denied!

You are not authorized to access the [Internal & External User Management] screen.

- Refresh the page or try again later.

- If the problem persists, contact your system administrator for assistance.

```
// Check State Valid
commonUtil.checkStateValid = function (stateName) {

    return '';

    if (stateName) {
        // New Popup
        if (stateName.indexOf('Pop.') > -1 ) {
            return '';
        }

        // Sample
        if (stateName.indexOf('sample.') > -1 ) {
            return '';
        }

        // Check only Sref Menu
        if (commonUtil.srefInfo && commonUtil.srefInfo[stateName]) {
            var dealerInfo = commonUtil.getDealerInfo();
            var userInfo = commonUtil.getUserInfo();
            var stateInfo = commonUtil.getMenuInfo(stateName);
            var stateTitle;
```

/ A welcome surprise

- 🚨 Users viewable!
- Note the create button...

Internal & External User Inquiry

[+ Create](#)

User Type:
Internal

User ID:
User ID

User Name:
User Name

Company Code:
All

User ID Active:
All

Email:
Email

[Search](#)

Search Results : 3781

[Print](#) [Excel](#)

Company Code	User Type	User ID	User Name	EMAIL	User ID Active
	Internal				No
	Internal				Yes
	Internal				Yes
	Internal				Yes
	Internal				Yes
	Internal			TMP_	No
	Internal			TMP_	No
	Internal			TMP_	No
	Internal			TMP_	No
	Internal				Yes

[First](#) [Previous](#) [1](#) [2](#) [3](#) [4](#) [5](#) ... [379](#) [Next](#) [Last](#)

/ The create user form

- Juicy access options
- Access Rights = super long list
 - Like a CVS receipt
- Moment of truth: submit...

Create Internal & External User

[Back to List](#) ☒ Reset

✓ Validation has been done! Click SUBMIT button to Save current record.

Account Information

* User Type: External User ID:

User ID Active: ☒

Contact Information

* Title: MR.

* First Name / MI: / * Last Name: TEST

* Email: * Confirm Email:

* Company Name: TEST COMPANY * Mobile: 123-456-1337

Access Information

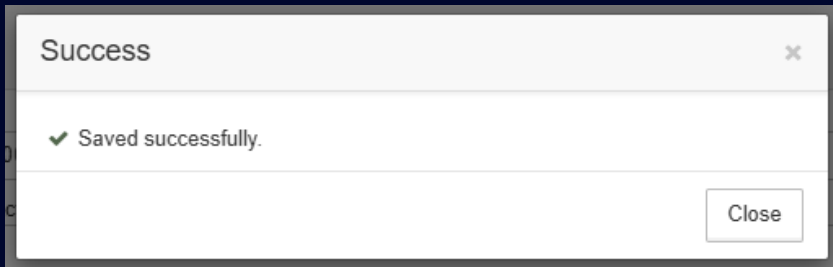
Access Level: ☒ National ☐ Region ☐ District ☐ Dealer

Accessible National List: ☒ Sales National ☒ Service National ☒ Parts National ☒ Finance National

Access Right

FUNCTIONAL GROUP					
System	Functional Group		Description	Assigned Menu	
	Parent	Child			
	Admin	Admin Dealer Administrators	The system Administrator responsible for setting up dealership.		☒
		Parts Order Entry/Submit	User can enter and submit Parts Orders.		☐
	Parts	Parts Return Entry/Submit	User can enter and submit Parts Returns.		☐

/ Congratulations! (for real)



Congratulations!

 has granted you access to the functional groups and access level listed below

SYSTEM ACCESS INFORMATION

	
User ID	
Temporary Password	
Authority(Funtional Group)	 Admin Dealer Administrators

/ Why it worked

- Create and retrieve users done via API call
- API server checked for valid JSESSIONID, but not if you were admin
- Not all APIs vulnerable, hence need to create national admin

/ The door is now open

- Account has access to **everything**
- First login: choose dealer to administer
- Let's dig in...

[illegible]

/ Doxxing you from the parking lot

- Time for a real-world example!
- Weekend project: go to Publix and find a target. Took seconds to find.



Not a Cybertruck

/ Doxxing you from the parking lot

- Swipe the VIN off the windshield real quick...
 - No amount of dirt can save you!



/ Doxxing you from the parking lot

- Consumer lookup/update tool exists in dealer system
- Supports lookup by VIN or first/last name
- Silent

Service / Consumer Information Update

Consumer Information Update

Page 1 / 1

[Back to List](#) ☒ Check Valid

Consumer Information

Vehicle Number			
Current Owner	* Last Name	First Name	Middle Initial
* Address		* City	
* State		* Zip Code	
Phone Number 1		N	Cell Phone
Phone Number 2			Extension
Email Address			Mail Flag Y
* Nearest Dealer		Q	Nearest Dealer Name



/ Hacking a car

- Many clever hacks to control cars (key fob, mobile apps, immobilizer)
- **The unexpected connection:** dealers have control over your car



/ Hacking a car

- Dealers have an enrollment systems to pair customers to their new car
- The pairing enables use of the mobile app
- Mobile app controls remote start and unlock functionality
- Pairing process could be abused to transfer ownership of cars

/ Hacking a car – prep

- Enlisted help of friend. Full consent obtained to test transfer
- I created fresh account on automaker's consumer website
- Test goal: switch ownership of car to my fresh account

/ Hacking a car – step 1

- Log in to enrollment system using dealer account
- Find my consumer account to kick off the process

1. Find or create an account

Account found

[Search again](#) [Create new account](#)

Eaton Zveare [Edit](#)

[Reset password](#)

Add vehicle

/ Hacking a car – step 2

- Obtain VIN. Can find on windshield
 - Remember: Consumer lookup tool supports lookup by name
- Enter VIN in form
- Pinky promise the request is legit 🤞

2. Enter VIN

Vehicle Identification Number (VIN)

⚠️

Check VIN

OR

Scan VIN

[Go Back](#)

This VIN is already connected to another account.
Adding this VIN to a new account will delete any connections to the existing account.
Confirm you have verified the customer's ownership based on one of the following:

1. The vehicle was just purchased at your dealership new or CPO
2. Current registration documents
3. Title
4. Other documents showing proof of ownership

Would you like to continue and remove the connection to the existing account?

Continue

[Cancel](#)

/ Hacking a car – step 3

- Grill the customer on their driving habits

3. Enter odometer and driving habits

Odometer Reading

Miles

1337

Driving Habits

Please select all of the customer's typical driving conditions and habits:

☐ Repeated short-distance driving (less than 5-10 miles).

☐ Extended periods of idling or low-speed operation.

☐ Extensive use of brakes.

☐ Driving on unpaved, gravel, muddy, rough, dusty or sandy roadways.

☐ Driving in areas where salt or other corrosive materials are used.

☐ Driving with heavy load or workload on the vehicle such as towing, mountain roads, or commercial use.

☐ Driving for a prolonged period in cold temperatures and/or extremely humid climates.

☐ More than 50% driving in heavy city traffic, during hot weather above 90°F (32°C).

☒ None of the above

Save And Continue

Cancel

/ Hacking a car – step 4

- Confirm your account info

4. Account info

Email

First Name

Eaton

Last Name

Zveare

Address

Address Line 2

City

State

▼

Zip

Primary

Mobile Number

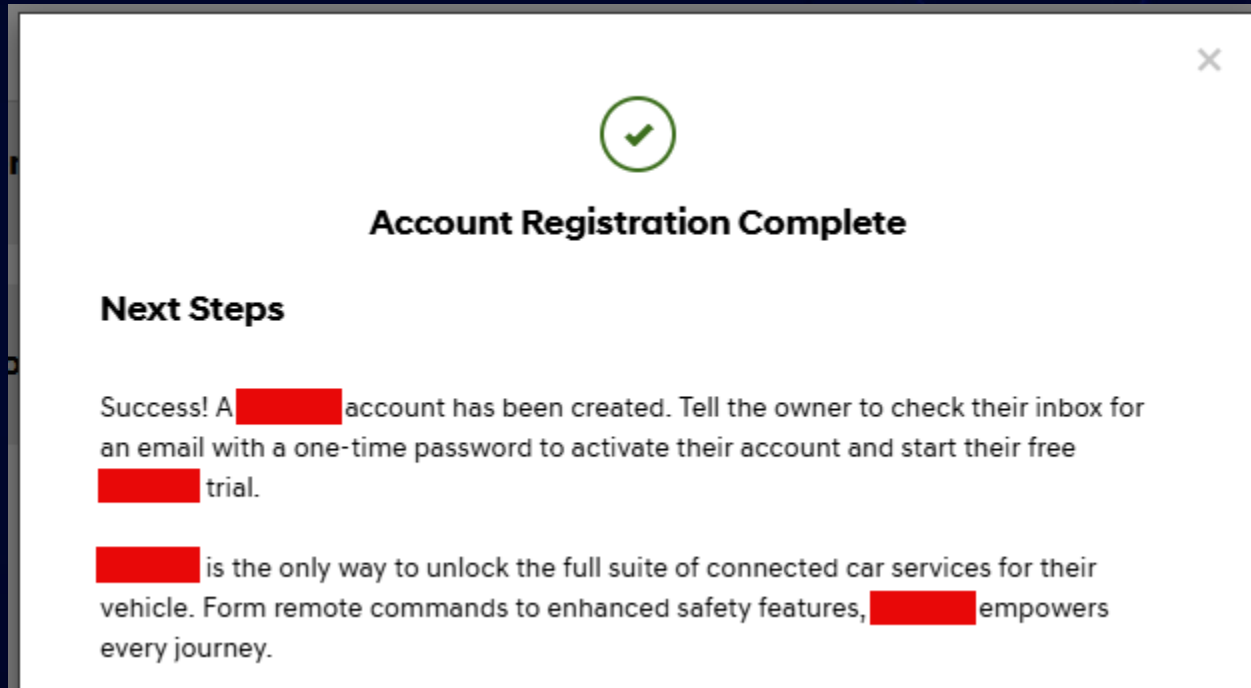
[Add Another Phone Number](#)

Save And Continue

[Cancel](#)

/ Hacking a car – success

- Ownership transfer complete



/ Hacking a car – control

- My account now owns the car
- To control it, mobile app is used
- Remote start, unlock, and location tracking confirmed working
- Car transferred back to friend without issue

/ Hacking a car – saying no

- Scenario: dealer offers to help set up your mobile app. You say no.
- Bad news: this makes things worse for you!
- Current owner gets email about transfer. You would **not** get it.
- Silent takeover of your car 🚗



/ Hacking a car – takeaways

- Victim gets email alert, but powerless to stop transfer
- Enrollment system relies on dealer employee trust
- No owner would have thought of this as backdoor
- Affected cars: MY12 onwards
 - Standard telematics module = affects most cars you see
- Only needed to know the first/last name of your target

/ Dealer System – SSO Control

- Dealer system comprised of many different websites
- Admin access to control SSO settings for them
- System ID will come in handy...

User Management

Menu Management

Permission Group Management

SSO Management

SSO Target System Management

Admin()

SSO Management

SSO Target System Management

SSO Source & Target System Inquiry

System Type

All

System ID

System Name

System Active

All

Belongs To

All

+ Create

Search

Search Results : 83

25

System Type	System ID	System Name	Target System	User Validation Type	Created By	System Active
Internal				All		YES
Internal				Token		YES
Internal				Token		YES
Extenal				Token		YES
Internal				Token		YES
Extenal				Token		YES
Internal				Token		YES
Extenal				Token		YES
Internal				Token		YES
Extenal				Token		YES

/ Dealer System – SSO settings in detail

User Management

Menu Management

Permission Group Management

SSO Management

SSO Target System Management

Admin() / SSO Management / SSO Target System Management

[Detail & Edit] SSO Source & Target System

[Back to List](#) [Reset](#) [Save](#) [Cancel](#)

System Basic Information

* System Type

Internal

* System ID

System Name

* Validation Duration(Sec)

120

* Validation ID

* Validation PW

System Active

☒

* Except Permission Check

Yes

* EP System Group ID

System Belongs To

Additional Properties

Default Redirect URL

Desktop URL

Mobile URL

* DEEP Link

No

User Validation

* Validation Type

All

* Allow Direct Access(For Dealer User)

Yes

Authorization Attributes

NO.

Attribute Name

Description

Active

Created By

8

CorporateCode

CorporateCode

☒

7

CorporateName

CorporateName

☒

6

DetailUserInfo

DetailUserInfo

☒

5

FirstName

FirstName

☒

4

IAMUserID

IAMUserID

☒

3

JobCode

JobCode

☒

2

LastName

LastName

☒

1

UserID

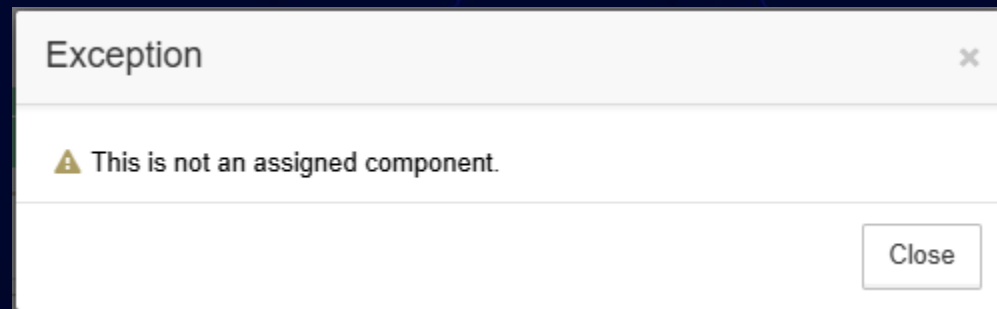
UserID

☒

[Add](#)

/ Sub-brand pivot

- Dealer luxury sub-brand uses same system
- Rebranded, different domain, different session ID
- National admin exploit did not work!
- Needed different way in...



Create account error

/ Sub-brand pivot – user impersonation

- Bombshell discovery: user impersonation feature
- Unlocked remaining systems
- Bypassed login & 2FA

Portal > Login As Dealer

Select a dealer user for single sign on to external sites.

Search Dealer

☐ By Dealer Code ☒ By Dealer Name

Dealer	Select
	☐
	☒
	☐

Select Role to Impersonate

- Business Development Center
- Business Development Center
- Business Development Center
- Business Development Center
- Business Development Center
- Customer Experience Manager
- Dealer Admin Support
- Dealer Admin Support

CONTACTS | SITE MAP | WEBSITE PRIVACY POLICY

/ Sub-brand pivot – SSO

- Dealer accounts can be associated with multiple dealerships
- Looked up dealers with combined stores
- Goal: find high-level person/account that worked for both

/ Sub-brand pivot – the process

- SSO login URL: *dealer-primary-brand.com/iam.gateway.AuthDispatcher?SSO_SYS_ID=XXX*
- When you load a dealer system, that is used to SSO you into it
- Many IDs. Of interest: the sub-brand ones
- **Steps:**
 1. Impersonate dealer user
 2. Replace *SSO_SYS_ID* with a sub-brand system ID
 3. Successfully logged in to sub-brand system as dealer user!
 4. With SSO session set, navigation to more sub-brand systems was possible

/ Sub-brand pivot – takeaways

- Failure to create national admin
- National admin account and user impersonation on main brand essential
- These cars hackable too

/ Access Review

- At this point, I have...
 - National admin access to every primary brand dealer
 - The ability to log in as (impersonate) any dealer account
 - Significant access to sub-brand dealer systems
- National admin could SSO into any system
- Let's explore some highlights!

/ TSD Dealer Admin

- *“Cloud-based, connected dealer solution that rises above the rest”*
- What you need to know: it's a courtesy/loaner platform
- Used when you get your car serviced



/ TSD Dealer Admin

Choose any dealer

The screenshot shows the 'LOCATIONS HUB' interface. At the top, there's a navigation bar with 'TSD/DEALER' and icons for ACCOUNT, HELP, TRAINING, and LOG OUT. Below this is a sub-navigation bar with links to Dashboard, Locations, Reports, Employees, Viewership, Hierarchy, OEM Roles, Management, and Safety Recall. A search bar for 'Search Locations' is present. The main content area is titled 'MY LOCATIONS' and shows a tree view on the left with 'Archive', 'CE', and 'CE1' categories. The right side displays a list of locations, each with fields for DEALER NUMBER, CONTACT, PHONE NUMBER, TSD NUMBER, EMAIL, and TIME ZONE. A 'VIEW LOCATION' button is visible for each entry.

Viewing loaner agreements

The screenshot shows the 'AGREEMENT HUB' interface. At the top, there's a navigation bar with 'TSD/DEALER' and icons for REPORTS, ACCOUNT, ADMIN, HELP, and TRAINING. Below this is a sub-navigation bar with links to Search Open, Appointment, Agreement, and Fleet. A search bar is present. The main content area is titled 'AGREEMENT HUB' and shows a list of loaner agreements. Each entry includes a date range, a 'Loaner' button, an 'Open' button, and fields for PHONE, LOCATION, and UNIT. A red box highlights the text 'Showing 1 to 100 of 4669 entries' and 'Just for 1 dealer'.

/ TSD Dealer Admin – Redacted PII



- National admin could not view PII for some reason

VIEW AGREEMENT ?

AGREEMENT: Open

CHECKOUT DATE: 01/29/2025 9:40 AM LOCATION: CLASSIFICATION: Service


Email Agreement


History



****, ****, ****

DATE OF BIRTH
December 31, 9999

DRIVER'S LICENSE
**** (****)

INSURANCE COMPANY

PHONE
+**** (Mobile) Invalid
Phone Number

EXPIRATION DATE
December 31, 9999

POLICY NUMBER

EMAIL

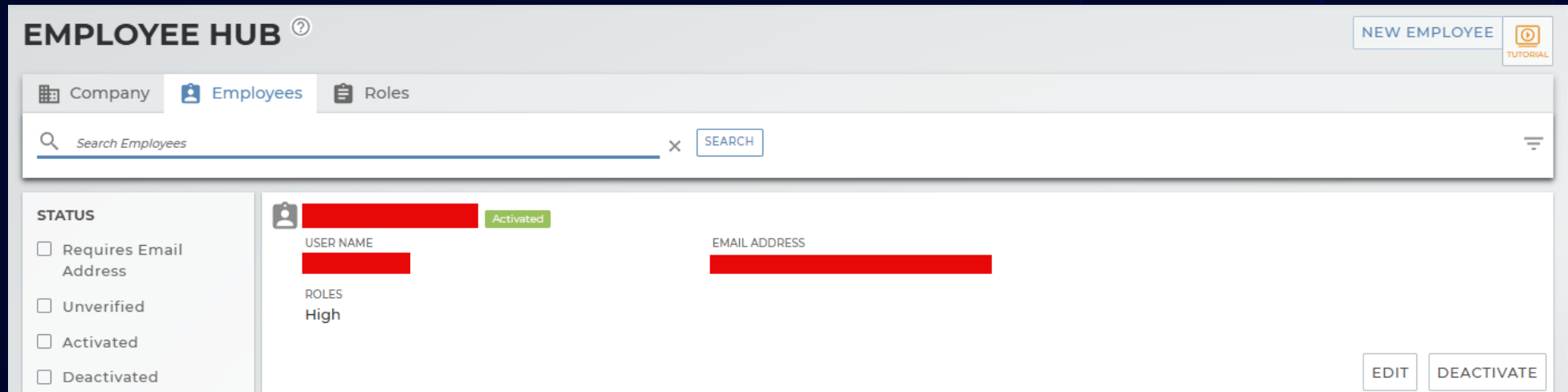
EXPIRATION DATE
December 31, 9999

FULL COVERAGE
No

No additional drivers.

/ TSD Dealer Admin – Impersonation

- Maybe PII can be viewed by specific dealer admins?
- Let's find one with “High” roles and try impersonating them...



The screenshot displays the 'EMPLOYEE HUB' interface. At the top, there are tabs for 'Company', 'Employees', and 'Roles'. A search bar labeled 'Search Employees' is present, along with a 'NEW EMPLOYEE' button and a 'TUTORIAL' icon. Below the search bar, a list of employees is shown. The first employee is highlighted, showing their status as 'Activated' and their role as 'High'. The 'USER NAME' and 'EMAIL ADDRESS' fields are redacted with red bars. On the left, there are filter checkboxes for 'Requires Email Address', 'Unverified', 'Activated', and 'Deactivated'. At the bottom right of the employee card, there are 'EDIT' and 'DEACTIVATE' buttons.

STATUS	USER NAME	EMAIL ADDRESS	ROLES
☐ Requires Email Address ☐ Unverified ☐ Activated ☐ Deactivated	[Redacted]	[Redacted]	High

/ TSD Dealer Admin – Roles

 Low				
PERMISSIONS				
▪ Access to Telematics Map	▪ Close Shuttle Agreement	▪ Flag Customer	▪ Reports	▪ View Customer Hub
▪ Allow Save Credit Card to Customer Profile	▪ Create New Appointment	▪ Inspection	▪ View Agreement Hub	▪ View Dashboard
▪ Close Agreement	▪ Create New Customer	▪ Open Agreement	▪ View Appointment Hub	▪ View Fleet Hub
	▪ Edit Customer	▪ Operations	▪ View Company Hub	▪ View Shuttle Agreement Hub
 Medium				
PERMISSIONS				
▪ Access to Telematics Map	▪ Create New Appointment	▪ Exchange Unit	▪ Override Minimum Age Restriction	▪ View Customer Hub
▪ Allow Opt-Out of Charges	▪ Create New Customer	▪ Flag Customer	▪ Performance	▪ View Dashboard
▪ Allow Save Credit Card to Customer Profile	▪ Edit Agreement	▪ Inspection	▪ Reports	▪ View Fleet Hub
▪ Cancel Appointment	▪ Edit Appointment	▪ Open Agreement	▪ View Agreement Hub	▪ View Reports
▪ Close Agreement	▪ Edit Company Information	▪ Open Shuttle Agreement	▪ View Appointment Hub	▪ View Shuttle Agreement Hub
▪ Close Shuttle Agreement	▪ Edit Customer	▪ Operations	▪ View Company Hub	▪ Write Reports
	▪ Edit Shuttle Agreement			
 High				
PERMISSIONS				
▪ Access to Telematics Map	▪ Create New Customer	▪ Edit Fleet	▪ Location Access Administration	▪ View Agreement Hub
▪ Add Custom Report	▪ Customer CSV Imports	▪ Edit Shuttle Agreement	▪ Manage Third Party Agreement Providers	▪ View Appointment Hub
▪ Add Fleet	▪ Customer Facing Configurations	▪ Edit System Report	▪ Open Agreement	▪ View Company Hub
▪ Agreement Void	▪ Display Alerts	▪ Edit Turnback Date	▪ Open Shuttle Agreement	▪ View Customer Hub
▪ Allow Modify Agreement Rate	▪ Edit Agreement	▪ Edit Turnback Miles	▪ Operations	▪ View Dashboard
▪ Allow Opt-Out of Charges	▪ Edit Appointment	▪ Enable Counter Bypass	▪ Override Minimum Age Restriction	▪ View Employee Hub
▪ Allow Save Credit Card to Customer Profile	▪ Edit Company Information	▪ Exchange Unit	▪ Performance	▪ View Fleet Hub
▪ Cancel Appointment	▪ Edit Custom Report	▪ Flag Customer	▪ Reminder Timing	▪ View Management Hub
▪ Close Agreement	▪ Edit Customer	▪ Fleet Import	▪ Reports	▪ View Roles Hub
▪ Close Shuttle Agreement	▪ Edit Dealership/Location Management	▪ Fleet Status	▪ Unvoid Agreement	▪ View Shuttle Agreement Hub
▪ Create New Appointment	▪ Edit Employee	▪ Inspection		

/ TSD Dealer Admin – PII Time!

- Workaround successful 

VIEW AGREEMENT

AGREEMENT:  [Open](#)

CHECKOUT DATE: 01/29/2025 9:40 AM LOCATION:  CLASSIFICATION: Service


Email Agreement


Exchange Unit


Close Agreement


Close




DATE OF BIRTH
September 27, 1993

PHONE
 (Mobile)

EMAIL


DRIVER'S LICENSE
 (New York)

EXPIRATION DATE
September 28, 2032

INSURANCE COMPANY
PROGRESSIVE

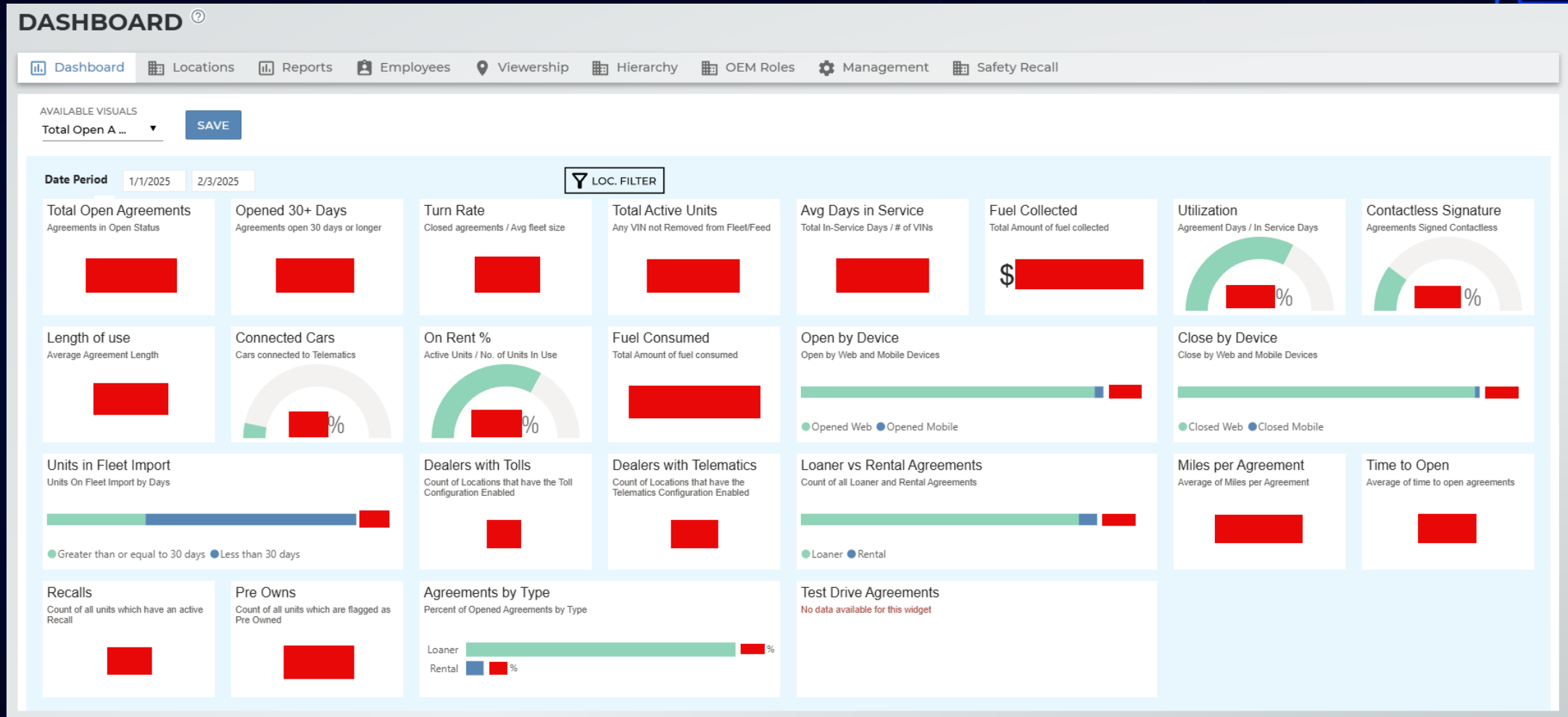
POLICY NUMBER


EXPIRATION DATE
December 8, 2026

FULL COVERAGE
No

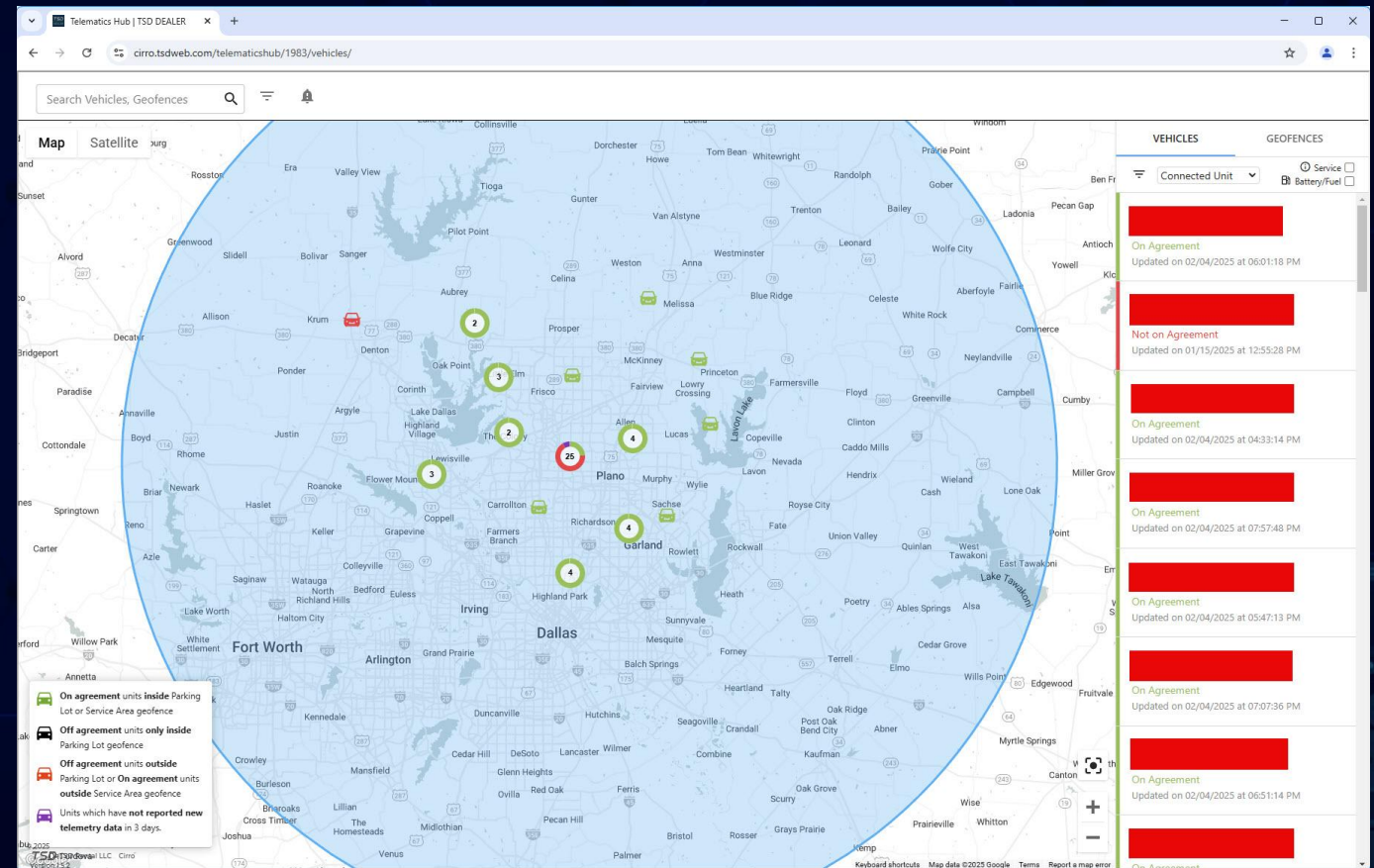
No additional drivers. [ADD DRIVER](#)

/ TSD Dealer Admin – Dashboard



/ TSD Dealer Admin – Connected Cars

- Optional telematics feature
- Real-time fleet tracking
- Prevents theft from customers
 - And geofence to ensure compliance



/ Dealer System – Employee PII

- Dealer employee SSN last 4 found in API response

The image displays a web application interface for 'Dealer User Inquiry' and a corresponding API response. The interface on the left shows search filters for Account Status, User ID, and User Name, with a search button. Below the filters, it shows 'Search Results : 171' and a table of results. The table has columns for User ID, Status, and User Name. The first row shows a 'Locked' status and a 'Y' in the User Name column. The second row shows an 'Active' status and a 'Y' in the User Name column. The third row shows an 'Active' status and a 'Y' in the User Name column. The fourth row shows an 'Active' status and a 'Y' in the User Name column. The fifth row shows an 'Active' status and a 'Y' in the User Name column. The sixth row shows an 'Active' status and a 'Y' in the User Name column. The seventh row shows an 'Active' status and a 'Y' in the User Name column. The eighth row shows an 'Active' status and a 'Y' in the User Name column. The ninth row shows an 'Active' status and a 'Y' in the User Name column. The tenth row shows an 'Active' status and a 'Y' in the User Name column. The eleventh row shows an 'Active' status and a 'Y' in the User Name column. The twelfth row shows an 'Active' status and a 'Y' in the User Name column. The thirteenth row shows an 'Active' status and a 'Y' in the User Name column. The fourteenth row shows an 'Active' status and a 'Y' in the User Name column. The fifteenth row shows an 'Active' status and a 'Y' in the User Name column. The sixteenth row shows an 'Active' status and a 'Y' in the User Name column. The seventeenth row shows an 'Active' status and a 'Y' in the User Name column. The eighteenth row shows an 'Active' status and a 'Y' in the User Name column. The nineteenth row shows an 'Active' status and a 'Y' in the User Name column. The twentieth row shows an 'Active' status and a 'Y' in the User Name column. 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The API response on the right shows a JSON object with a 'result' field containing a 'retData' object. The 'retData' object has a 'ssn' field with the value '1234', which is highlighted in red. A red arrow points to this value with the text 'SSN last 4'.

```
{
  "result": {
    "retMsg": "",
    "retData": {
      "ssn": "1234"
    },
    "retCd": "S"
  }
}
```

/ Dealer System – Employee PII

- Viewing full SSN possible, but action logged
- Not attempted, but likely to work

Please Enter The Reason For Inquiring 'Personnel Enrollment Info'.

Reason:

Verify employee's SSN
Verify employee's address

Close

/ Dealer System – Audit Log

- Can view that log, of course 🚗

Admin() / User Management / Audit Report

Audit Report

Viewer Viewee

User Type: User ID: User Name: Reason:

Created Date: - Company / Dealer code:

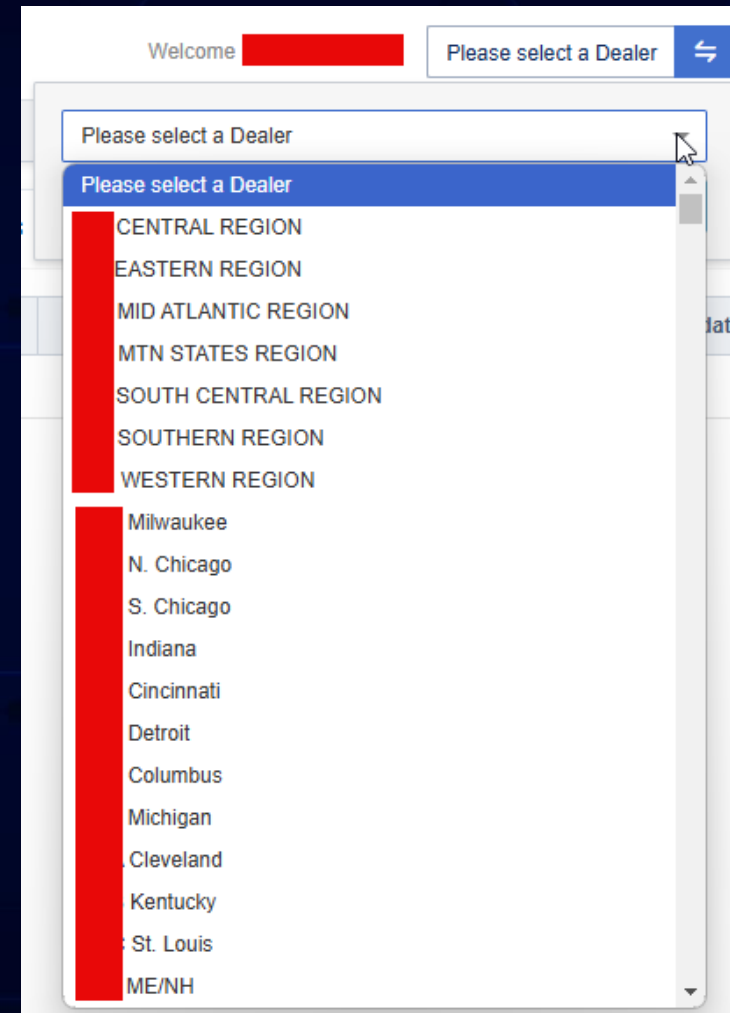
Search Results : 1000

25

User ID	User Name	Company / Dealer	Count
			2
			5
			1
			2
			1
			4

/ Dealer System – Inventory Control

- Dealers have allocation/order tool
- Place orders with HQ, dealer trade
- National admin had total control over it



/ Dealer System – Inventory Control Detail

CRITERIA FILTERS Input VIN / AON All ETA Dates All Model Group All Years All Models All Packages All Exterior Colors All Interior Colors All Vehicle Status ADDITIONAL FILTERS SEARCH / REFRESH RESET FILTERS	SUMMARYDETAILDEALER PIPELINE REPORTDEALER ETA REPORTSRC MARK OPENMARK SOLDREMOVE OPEN / SOLDPIPELINE TRADEPRINTEXPORTFREIGHT VERIFY <table><tr><th></th><th>AON or VIN</th><th>MY</th><th>Model</th><th>Description</th><th>Package</th><th>Generic Color</th><th>Color</th><th>Accessories</th><th>Allocation</th><th>MSRP</th></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>BLACK</td><td></td><td></td><td>241001</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>GRAY</td><td></td><td></td><td>241003</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>GRAY</td><td></td><td></td><td>241103</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>RED</td><td></td><td></td><td>2410M0</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>GRAY</td><td></td><td></td><td>2410M0</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>BLACK</td><td></td><td></td><td>2410M0</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>WHITE</td><td></td><td></td><td>2410M0</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>WHITE</td><td></td><td></td><td>2410M0</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>GRAY</td><td></td><td></td><td>241003</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>WHITE</td><td></td><td></td><td>241003</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>BLUE</td><td></td><td></td><td>241003</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>RED</td><td></td><td></td><td>241003</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>GRAY</td><td></td><td></td><td>241003</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>GRAY</td><td></td><td></td><td>2410PJ</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>SILVER</td><td></td><td></td><td>2410PJ</td><td>TBI</td></tr><tr><td>☐</td><td></td><td>2025</td><td></td><td></td><td>01 - Standard</td><td>WHITE</td><td></td><td></td><td>2410PJ</td><td>TBI</td></tr></table> 2001 - 200 of 1986 items 12345678910											AON or VIN	MY	Model	Description	Package	Generic Color	Color	Accessories	Allocation	MSRP	☐		2025			01 - Standard	BLACK			241001	TBI	☐		2025			01 - Standard	GRAY			241003	TBI	☐		2025			01 - Standard	GRAY			241103	TBI	☐		2025			01 - Standard	RED			2410M0	TBI	☐		2025			01 - Standard	GRAY			2410M0	TBI	☐		2025			01 - Standard	BLACK			2410M0	TBI	☐		2025			01 - Standard	WHITE			2410M0	TBI	☐		2025			01 - Standard	WHITE			2410M0	TBI	☐		2025			01 - Standard	GRAY			241003	TBI	☐		2025			01 - Standard	WHITE			241003	TBI	☐		2025			01 - Standard	BLUE			241003	TBI	☐		2025			01 - Standard	RED			241003	TBI	☐		2025			01 - Standard	GRAY			241003	TBI	☐		2025			01 - Standard	GRAY			2410PJ	TBI	☐		2025			01 - Standard	SILVER			2410PJ	TBI	☐		2025			01 - Standard	WHITE			2410PJ	TBI
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/ Dealer System – Inventory Control (regional)

- Special regional access

CRITERIA FILTERS

Accepted Vehicles

All Dealers

All Allocation Group

All Years

All Models

All Packages

All Colors

ADDITIONAL FILTERS

+

SEARCH / REFRESH

RESET FILTERS

ALLOCATION LIST

REGION DETAIL

REGION FINALIZE

ALLOCATION STATISTICS

PORT OPTIONS

TURNDOWN

PRINT

EXPORT

	Line #	District	From Dealer	To Dealer	Model Year	Model	Description	Package	Generic Color	Color	Accessories	Modified	F
☐	1	MA1			2025			01-Standard					
	2	MA1			2025			01-Standard					
	9999	MA2			2025			01-Standard					
	9999	MA2			2025			01-Standard					
	9999	MA2			2025			01-Standard					
	9999	MA2			2025			01-Standard					
	5	MA2			2025			01-Standard					
	6	MA2			2025			01-Standard					
	9999	MA2			2025			01-Standard				T	
	1	MA1			2025			01-Standard					
	2	MA1			2025			01-Standard					
	3	MA1			2025			01-Standard					

200 1 - 200 of 4110 items

1

2

3

4

5

6

7

8

9

10

...

/ FreightVerify – Track cars being shipped...

Search Results - FreightVerify

portal.freightverify.com/vinsearch

Search Results

Search VIN

VIN

Search

Advanced Search

Origin Destination Carrier Product Type VIN Status VIN Exception Last Milestone Pickup Date

Delivery Date Order Type Partner Shipment Status Shipment Exceptions Origin Country Origin Region

Destination Country Destination Region Dealer Zone Dealer District Current Location Route ID

Batch Search

454,291 Total VINs

Reset Save this search Download Data (.csv)

Watch	VIN	Last Milestone	Last Update	Origin Details	Destination Details	Active Exceptions
☆	[REDACTED]	Vehicle Received (RT) Event Time: 11:47 EST 10/02/2024	At Location Port of Tacoma - USTIW (USTIW) Tacoma, WA Source : [REDACTED] 04:13 EST 01/27/2025 Route ID - N/A	Actual Pickup 17:40 EST 09/09/2024	Actual Delivery 11:00 EST 09/24/2024 ETA: TBD	
☆	[REDACTED]	Railcar Spotted (SPT) Event Time: 05:33 EST 10/07/2024	At Location Source - Milestone 05:33 EST 10/07/2024 Route ID - N/A	Actual Pickup 11:08 EST 09/19/2024	Actual Delivery 05:32 EST 10/07/2024 ETA: TBD	
☆	[REDACTED]	Railcar Spotted (SPT) Event Time: 05:33 EST 10/07/2024	At Location Source - Milestone 05:33 EST 10/07/2024 Route ID - N/A	Actual Pickup 11:08 EST 09/19/2024	Actual Delivery 05:32 EST 10/07/2024 ETA: TBD	
☆	[REDACTED]	Railcar Spotted (SPT) Event Time: 05:33 EST 10/07/2024	At Location Source - Milestone 05:33 EST 10/07/2024 Route ID - N/A	Actual Pickup 11:08 EST 09/19/2024	Actual Delivery 05:32 EST 10/07/2024 ETA: TBD	
☆	[REDACTED]	In Production (MA) Event Time: 19:16 EST 12/06/2024	At Location Costa Mesa, CA Source : [REDACTED] 11:30 EST 01/24/2025 Route ID - N/A	MA [REDACTED] Montgomery, AL	[REDACTED] FORT MYERS, FL ETA: Pending Dispatch	
☆	[REDACTED]	At Port (PA) Event Time:	At Location Port Hueneme (HU) Port Hueneme Naval CBC, CA	HU Port Hueneme	[REDACTED]	

Previous Page 1 of 22715 20 rows Next

Powered by FREIGHTVERIFY

sample [REDACTED] Log Out

... and cancel them!

The screenshot displays the 'Shipment Detail' page in the FreightVerify portal. The interface includes a sidebar with navigation options like 'Shipments', 'Finished Vehicle', 'Business Intelligence', 'Administration', and 'Training & Resources'. The main content area features a map of the Eastern United States with a blue line indicating a rail route from Chicago to Philadelphia. A red circle with an 'A' is located near Atlanta, and a red circle with an 'R' is near New York. To the right of the map, a 'Details' panel lists shipment information: Shipment ID, Mode (Rail), Shipper, Carrier (CSX TRANSPORTATION INC), SCAC (CSXT), Current Road (CSX TRANSPORTATION INC), Trailer/Equipment #, Train ID (CSXT), Loaded Status, and Loaded. An 'Actions' dropdown menu is open, showing options: Alert Me, Assign Asset/Trailer, Unassign Asset, Report Arrival/Departure Events, Report Destination Arrival Estimate, Add Bill of Lading/Manifest Numbers, and Cancel Shipment. A red arrow points to the 'Cancel Shipment' option. Below the details panel, the 'Arrivals and Departures' section shows the origin as a redacted box and the destination as 'Port of Philadelphia - Navy Yard'. It also displays scheduled arrival windows: 09:03 EST to 09:03 EST on 01/09/2025, and 23:00 EST to 23:00 EST on 01/21/2025. The user 'sampl' is logged in, and the page is powered by FreightVerify.

/ Finance Portal

- When you finance a car or protection plan, this is used
- Access to customer personal info via PDF links
- You had a contract with the dealer? Your info was ripe for the taking!

So many! 

/ Finance Portal – Contract

PROTECTION PLAN			
[REDACTED]			
CUSTOMER INFORMATION			
Customer Name [REDACTED]		Co-Customer Name [REDACTED]	
Address [REDACTED]		Address [REDACTED]	
City LITHIA	State FL	ZIP 33547	City LITHIA
Email [REDACTED]	(Area Code) Telephone Number [REDACTED]	Email [REDACTED]	(Area Code) Telephone Number [REDACTED]
COVERED VEHICLE INFORMATION			
Year 2024	Make [REDACTED]	Model [REDACTED]	Manufacturer's Original In-Service Date 12/09/2024
VIN [REDACTED]	Odometer Reading on Purchase Date 14		Covered Vehicle Purchase/Lease Date
SELLING DEALER INFORMATION			
Dealer Name [REDACTED]		Dealer Number [REDACTED]	Phone [REDACTED]
Address [REDACTED]	City [REDACTED]	State [REDACTED]	ZIP [REDACTED]
Lienholder/Lessor Name [REDACTED]		Lienholder/Lessor Address PO BOX 30205 [REDACTED]	
F&I Manager [REDACTED]			
INFORMATION			
Purchase Price 3,499.00		Purchase Date 12/09/2024	
COVERAGE TYPE	USAGE INTERVAL	OIL TYPE	TERM
BASIC If nothing is listed above, Basic Maintenance will apply.	NORMAL If nothing is listed above, the Normal Usage Interval will apply.	SYNTHETIC If nothing is listed above, the Conventional Oil option will apply. The Synthetic Oil option is only available for the Normal Usage Interval.	YEARS*/ [REDACTED] MILES* *The Term in Years is measured from the Manufacturer's Original In-Service Date. Mileage is based on the number of miles driven from zero, not the number of miles driven since the Maintenance Wrap Purchase Date.
THE PURCHASE OF THIS [REDACTED] AGREEMENT IS VOLUNTARY AND NOT REQUIRED IN ORDER TO OBTAIN FINANCING OR TO PURCHASE THE COVERED VEHICLE.			
WITH MY SIGNATURE BELOW, I VOLUNTARILY ELECT TO PURCHASE THIS [REDACTED] AGREEMENT FOR THE PURCHASE PRICE SHOWN ABOVE.			
Customer's Signature [REDACTED]		Date 12/09/2024	
Co-Customer's Signature [REDACTED]		Date 12/09/2024	
Selling Dealer's Signature [REDACTED]		Date 12/09/2024	
[REDACTED]			

/ Optimum Auto Solutions Portal

- Optimum Auto Solutions makes dealer/franchise management platform
- Used to measure performance of dealers, KPIs, finances, etc
- National access = massive trove of internal info and metrics

/ Optimum Portal – Finances

Reports

Search

Financial Statement

Performance Management +

Profit Tempo

Dealership Overview +

New Vehicle Dept +

Used Vehicle Dept +

Service Dept +

Parts Dept +

Dealer PCA Analysis Report

Dealer Profit Performance Report

DASHBOARDS

REPORTS

ANALYSIS TOOLS

DEALER

SUBMISSIONS

SUPPORT

EN-US

TEST (NTL)

National Users

Financial Statement

Network Group Dealer

Report Parameters

Network: NTL - NATIONAL | Value: Average | Period: Dec-24

PAGE 1 PAGE 2 PAGE 3 PAGE 4 PAGE 5 PAGE 6 PAGE 7

ENTITY NAME		NATIONAL			
2024 FINANCIAL STATEMENT		ENTITY CODE		COVERING PERIOD FROM: Dec 24 THRU Dec 24	
ASSETS		ACCT NO	AMOUNT OMIT CENTS	LINE NO	LIABILITIES
CURRENT ASSETS		NO			CURRENT LIABILITIES
Petty Cash		200		1	ACCOUNTS PAYABLE - Trade Creditors
Cash In Bank		202		2	Service, P&A, Body Shop Credit Balances
Cash On Hand		203		3	License & Registration Fees
Contracts in Transit		205		4	Customer Deposits Contract Units: 623
Securities, Money Market & CD's		260		5	Vehicle Protection & Warranties Payable
TOTAL CASH AND CONTRACTS (Lines 1 to 5)				6	Vehicle Lien Payable
RECEIVABLES				7	NOTES PAYABLE
Customer Notes (Past Due)		210		8	New Vehicles & Demonstrators
Customer Accts, Service, P&A, BS (Past Due)		220A		9	New Vehicles and Demonstrators - Other Franchise
Accounts Receivable - Retail Vehicles		222A		10	Used Vehicles -
Lease & Rental Accounts		223		11	Used Vehicles - Other
Wholesale & Dealer Trade		224		12	Lease and Rental Units
Service Contract Receivables		225		13	Other
Transportation Claims		229A		14	Current Portion -Long Term Debt
Trade Receivables Affiliates		261		15	TOTAL ACCOUNTS & NOTES PAYABLE (Lines 1 to 14)
Finance & Insurance Receivables		264		16	ACCRUED LIABILITIES
Allowance for Doubtful Accounts		340		17	Interest
NET CUSTOMER RECEIVABLES (Lines 8 to 17)				18	Payroll
FACTORY RECEIVABLES				19	Insurance
Warranty Claims/P.D.I. (Past Due)		226A		20	Taxes-Payroll
Warranty Claims/P.D.I. Other (Past Due)		226B		21	Taxes-Sales
Other Factory Receivables/Incentives (Past Due)		227A		22	Taxes-Other than P/R, Sales & Income Tax
Other Factory Receivables/Incentives-Other (Past Due)		227B		23	Income Taxes-Previous Year
Holdback - (Past Due)		228A		24	Income Taxes-Current Year
Holdback - Other (Past Due)		228B		25	Bonuses-Employees
Floor Plan Assistance - (Past Due)		229		26	Bonuses-Owners
Floor Plan Assistance - Other Franchise (Past Due)		229B		27	Pension Fund / Profit Sharing
Wholesale Assistance - (Past Due)		233		28	Other
TOTAL FACTORY RECEIVABLES (Lines 20 to 28)				29	Reserve for Reposs. F & I, Sevice Contract Losses
Due from Finance Companies		262		30	TOTAL ACCRUED LIABILITIES (Lines 17 to 29)
Account Payable Debit Balances		300		31	TOTAL CURRENT LIABILITIES (Lines 15 + 30)
TOTAL RECEIVABLES (Lines 18 + 29 + 30 + 31)				32	Lease and Rental Units
INVENTORIES				33	Deferred Income Taxes
Demonstrators: (Units) 2		230A		34	Long Term Debt
Demonstrators - Other (Units) 0		230B		35	Mortgages Payable
New Vehicles (Units) 100		231A		36	Notes Payable - Officers & Owners

/ Optimum Portal – KPIs

- Key Performance Indicator
- Measure of performance

NCM KPI Report							
Report Parameters							
National: NTL - NATIONAL Process With: Network Data Value: MTD Period: Dec-24							
	MTD						
	NTL - NATIONAL	CE - CENTRAL	EA - EASTERN	MA - MID ATLANTIC	MS - MOUNTAIN STATES	SC - SOUTH CENTRAL	SO - SOUTHERN
NCM - Return on Sales - Benchmark:							
NCM - Net Profit to Gross - Benchmark:							
NCM - Fixed Absorption - Benchmark:							
NCM - Used to New Retail Sold Ratio - Benchmark:							
New Vehicle							
NCM - New Vehicle Retail Gross per Unit after Incentives (GP-PNVR) - Benchmark:							
NCM - New F&I (& Other Franchise) Income PNVR - Benchmark:							
NCM - New Salesperson Compensation % Retail Gross - Benchmark:							
NCM - New Vehicle Salaries & Wages % Dept Gross - Benchmark:							
NCM - New Vehicle Net Advertising % Retail Gross - Benchmark:							
NCM - New Vehicle Operating Profit % Dept Gross - Benchmark:							
NCM - New Vehicle Dept Gross per New Employee Avg Month - Benchmark:							
NCM - New Dept Gross Profit per Salesperson Avg Month - Benchmark:							
NCM - New Retail Units per Salesperson Avg Month - Benchmark:							
NCM - New Vehicle Inventory Days Supply-Cost - Benchmark:							
Used Vehicle							
NCM - Used Vehicle Retail Gross PUVR - Benchmark:							
NCM - Used Wholesale Gross Profit per Unit Wholesaled - Benchmark:							
NCM - Used F&I Income PUVR - Benchmark:							
NCM - Used Sales Compensation % of Retail Gross - Benchmark:							
NCM - Used Vehicle Dept Salaries & Wages % of Dept Gross - Benchmark:							
NCM - Used Net Advertising % of Retail Gross - Benchmark:							
NCM - Used Operating Profit % Dept Gross - Benchmark:							
NCM - Used Dept Gross per Used Employee Avg Month - Benchmark:							
NCM - Used Dept Gross per Salesperson Avg Month - Benchmark:							
NCM - Used Retail Units per Salesperson Avg Month - Benchmark:							
NCM - Used Vehicle Inventory Days Supply - Cost - Benchmark:							
Service							
NCM - Service Customer Labor Gross % Sales - Benchmark:							
NCM - Service Customer Sales per RO - Benchmark:							
NCM - Service Hours Per RO (All Labor Types) - Benchmark:							
NCM - Service Hours per Customer RO - Benchmark:							

/ Optimum Portal – Dealer Rankings

- Ever wanted to know the real good/bad dealers?

KPI Ranking

Report Parameters

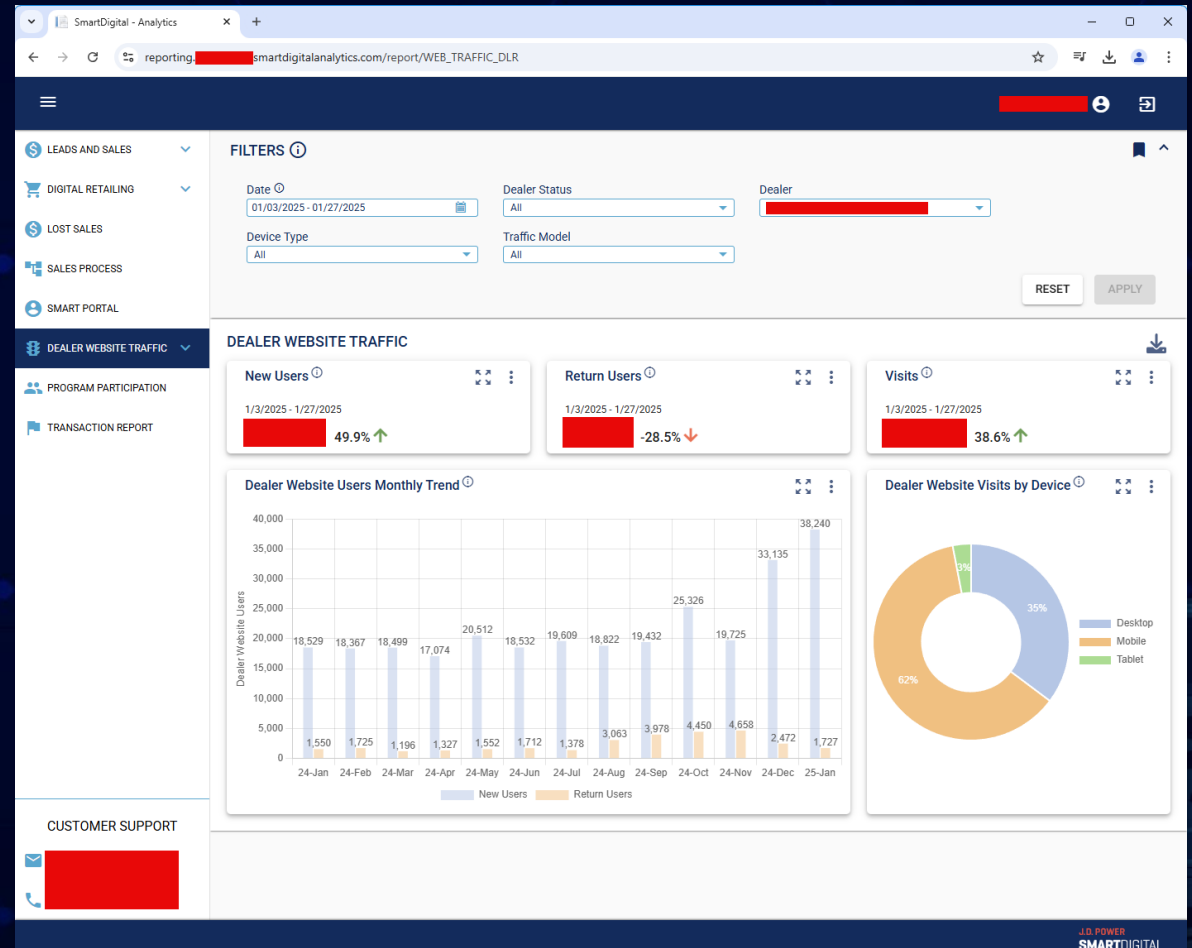
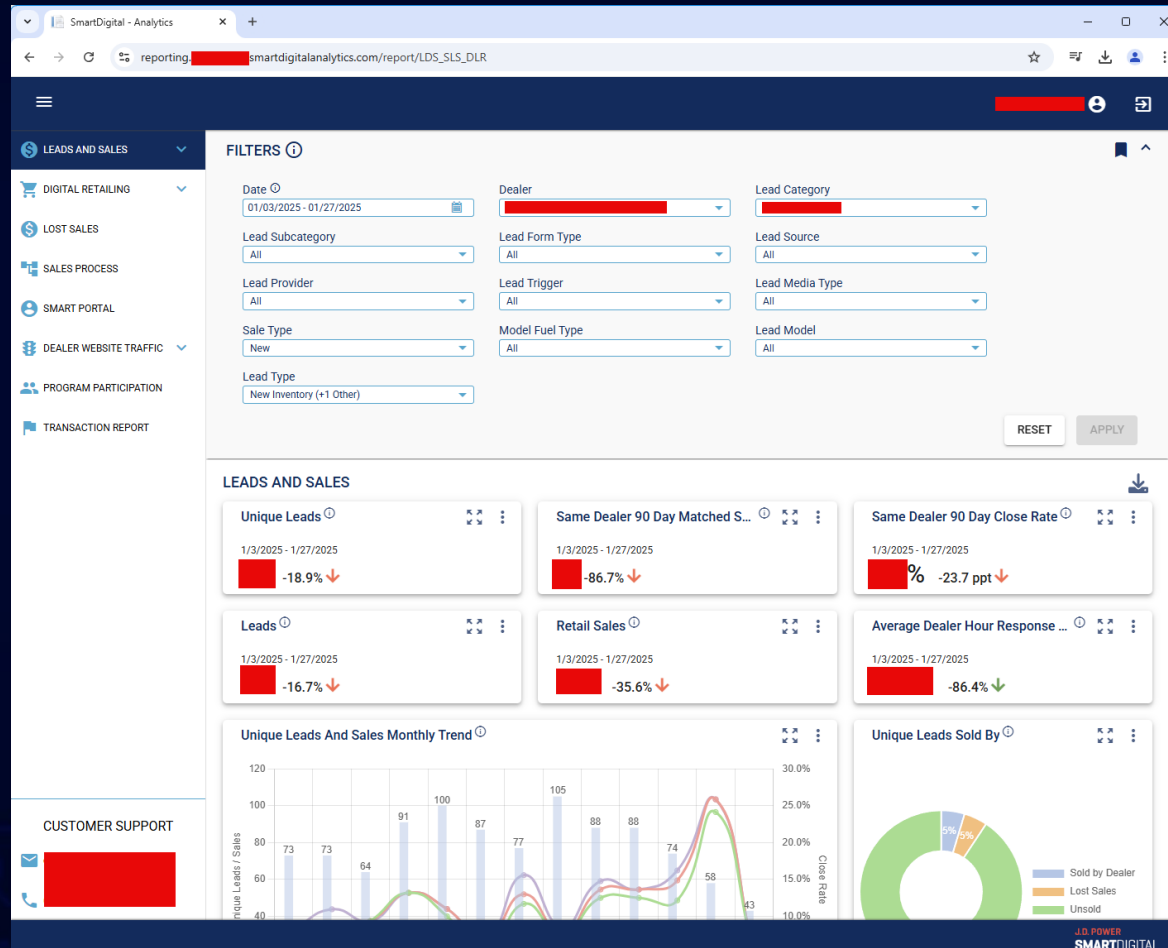
Network: NTL - NATIONAL | Period: Dec-24 | All Dealers | Chart Value: YTD

Return On Sales Total Dealership			Net-to-Gross Total Dealership		
Network	YTD	Rank	Network	YTD	Rank
	42.98%	1		1,688.53%	1
	26.28%	2		296.31%	2
	12.06%	3		88.69%	3
	10.16%	4		84.64%	4
	9.49%	5		80.42%	5

	-8.38%	834		-102.27%	834
	-9.16%	835		-103.92%	835
	-9.35%	836		-111.11%	836
	-10.69%	837		-115.89%	837
	-11.18%	838		-120.09%	838
	-12.40%	839		-132.43%	839
	-15.10%	840		-142.09%	840
	-17.72%	841		-174.71%	841
	-19.77%	842		-633.41%	842
NTL - NATIONAL	2.37%	Out of 847 in NTL	NTL - NATIONAL	20.49%	Out of 847 in NTL

/ J.D. Power Smart Analytics

- For monitoring website analytics



/ J.D. Power Lead Portal

- The salesperson's best friend

The screenshot shows the 'Lead Viewer' section of the J.D. Power Lead Portal. The interface includes a navigation bar with 'SmartPortal' and 'SmartAnalytics' tabs. The main header displays 'AUTODATA SOLUTIONS' and a welcome message. Below the header, there are filters for 'Lead Class' (Sales Lead), 'From (Received Date)' (01-Jan-2024), and 'To (Received Date)' (28-Jan-2025). An 'Advanced Search' button is also present. The results section shows 35,434 results, with a download option for CSV, PDF, or XLSX. A table of leads is displayed with columns for Indicator, Id, Date, Dealer ID, Customer, Type Make Model, Price, Lead Source - Lead Program, Request Status, and Download. The table lists four leads with their respective details.

Indicator	Id	Date	Dealer ID	Customer	Type Make Model	Price	Lead Source - Lead Program	Request Status	Download
VS_5434	01/28/2025 07:56 PM EST				NEW	\$0	Truecar/U.S. News -- Dealer Website Non-Endorsed		
VS_5012	01/28/2025 07:44 PM EST				NEW	\$0	Edmunds -- Dealer Website Non-Endorsed		
TG_3972	01/28/2025 07:13 PM EST				USED	\$22	J.D. Power SmartLeads -- TopAutoLeads	Request Credit	
VS_3556	01/28/2025 07:00 PM EST				NEW	\$0	Truecar/Truecar -- Dealer Website Non-Endorsed		

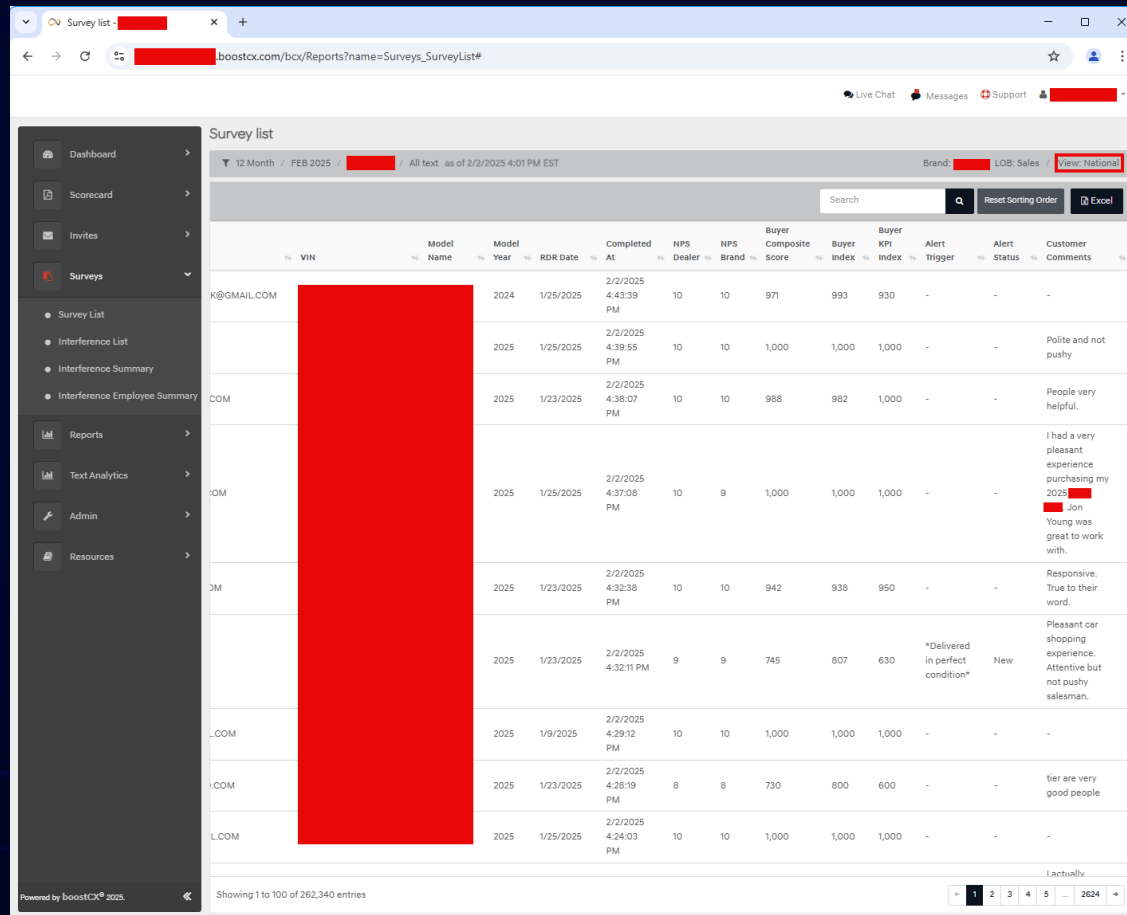
The screenshot shows the 'Lead Details' section of the J.D. Power Lead Portal. The interface includes a navigation bar with 'SmartPortal' and 'SmartAnalytics' tabs. The main header displays 'AUTODATA SOLUTIONS' and a welcome message. Below the header, there are tabs for 'Lead Details' and 'Disposition Details'. The 'Lead Details' tab is active, showing a list of fields and their values. The 'Disposition Details' tab is also visible, showing a table of events.

Field	Value
Document ID	
Lead ID	
Dealer	
Lead Received Timestamp	01/28/2025 07:56 PM EST
Lead Delivered Timestamp	01/28/2025 07:56 PM EST - to CRM: VinSolutions
CRM Event Delivery TimeStamp	01/28/2025 08:03 PM EST
Lead Response (Dealer Hours) (HH:mm)	0:00
Lead Source	Truecar/U.S. News
Lead Program(s)	Digital Dealer (Dealer Website Non-Endorsed)
Customer Name	
Customer Email	
Customer Phone	
Customer Address	
Vehicle Details	NEW 2025
Trim Name	FWD
Color Preference Exterior / Interior	N/A
Best Time to Contact	Cell Phone
Comments	*** DEALER PORTAL *** https://dealerportal.truecar.com/dfe/prospects/ ?xt=1 *** PROSPECT COMMENTS *** Consumer selected *** SEARCH ZIP *** Search ZIP is 32948 which is different than the Address ZIP. Search ZIP was 32948, member is aware that they are 133 miles away and still selected your dealership.

Sender	CRM/ DRS TimeStamp	Event Time	Event Code	Event Description	Comments
VIN	01/28/2025 08:03 PM EST	01/28/2025 08:01 PM EST	CUST_CONTACTED_PHONED	Salesperson responded to the prospect via phone (auto-responder will not trigger this event)	
VIN	01/28/2025 08:03 PM EST	01/28/2025 07:57 PM EST	CUST_CONTACTED_PHONED	Salesperson responded to the prospect via phone (auto-responder will not trigger this event)	
VIN	01/28/2025 08:03 PM EST	01/28/2025 07:56 PM EST	CUST_CONTACTED_EMAILED	Salesperson responded to the prospect via email (auto-responder will not trigger this event)	
VIN	01/28/2025 08:03 PM EST	01/28/2025 07:55 PM EST	LEAD_OPENED	Salesperson opened the lead for review	
VIN	01/28/2025 08:03 PM EST	01/28/2025 07:54 PM EST	AUTO_ASSIGNED	CRM System sent an automatic email or phone response to the prospect	

/ boostCX Surveys

- Those surveys we all love. Want to see what your friend really thinks of their car? I could find out!



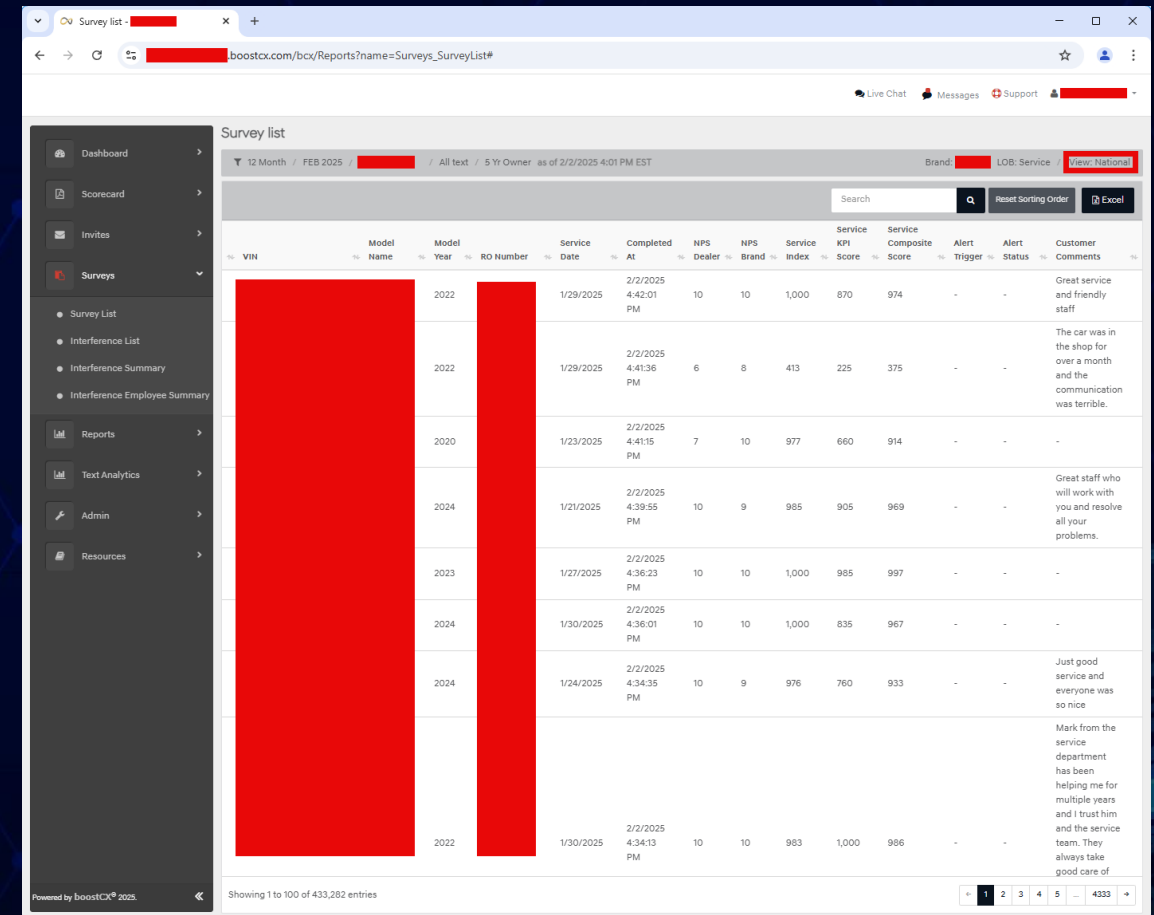
Survey list

12 Month / FEB 2025 / [Redacted] / All text as of 2/2/2025 4:01 PM EST

Brand: [Redacted] LOB: Sales View: National

VIN	Model Name	Model Year	RDR Date	Completed At	NPS Dealer	NPS Brand	Buyer Composite Score	Buyer Index	Buyer KPI Index	Alert Trigger	Alert Status	Customer Comments
K@GMAIL.COM	[Redacted]	2024	1/25/2025	2/2/2025 4:43:39 PM	10	10	971	993	930	-	-	-
[Redacted]	[Redacted]	2025	1/25/2025	2/2/2025 4:39:55 PM	10	10	1,000	1,000	1,000	-	-	Polite and not pushy
COM	[Redacted]	2025	1/23/2025	2/2/2025 4:38:07 PM	10	10	988	982	1,000	-	-	People very helpful.
OM	[Redacted]	2025	1/25/2025	2/2/2025 4:37:08 PM	10	9	1,000	1,000	1,000	-	-	I had a very pleasant experience purchasing my 2025 [Redacted] Young was great to work with.
OM	[Redacted]	2025	1/23/2025	2/2/2025 4:32:38 PM	10	10	942	938	950	-	-	Responsive. True to their word.
[Redacted]	[Redacted]	2025	1/23/2025	2/2/2025 4:32:11 PM	9	9	745	807	630	*Delivered in perfect condition*	New	Pleasant car shopping experience. Attentive but not pushy salesman.
.COM	[Redacted]	2025	1/9/2025	2/2/2025 4:29:12 PM	10	10	1,000	1,000	1,000	-	-	-
.COM	[Redacted]	2025	1/23/2025	2/2/2025 4:28:19 PM	8	8	730	800	600	-	-	tier are very good people
.COM	[Redacted]	2025	1/25/2025	2/2/2025 4:24:03 PM	10	10	1,000	1,000	1,000	-	-	-

Showing 1 to 100 of 262,340 entries



Survey list

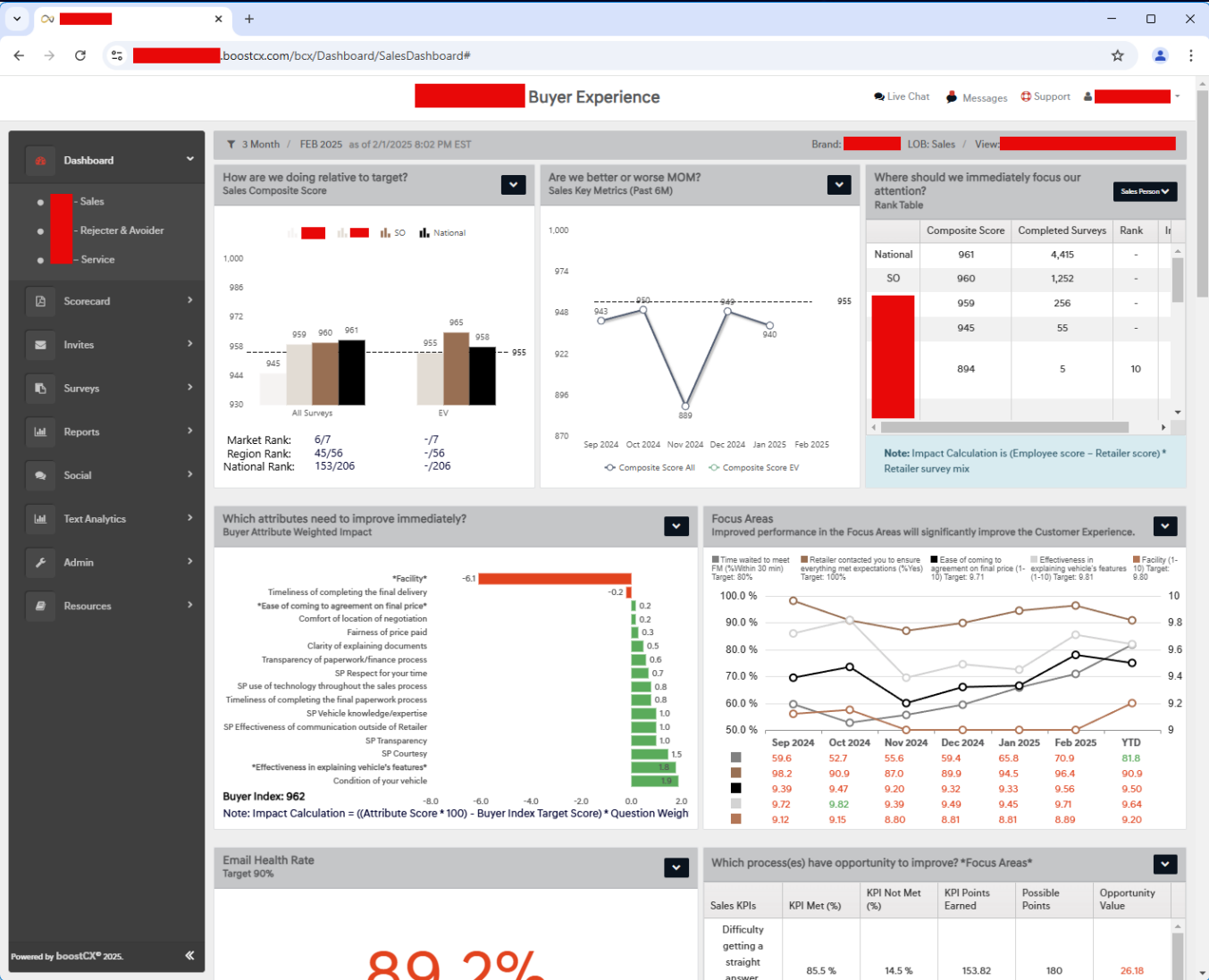
12 Month / FEB 2025 / [Redacted] / All text as of 2/2/2025 4:01 PM EST

Brand: [Redacted] LOB: Service View: National

VIN	Model Name	Model Year	RO Number	Service Date	Completed At	NPS Dealer	NPS Brand	Service Index	Service KPI Score	Service Composite Score	Alert Trigger	Alert Status	Customer Comments
[Redacted]	[Redacted]	2022	[Redacted]	1/29/2025	2/2/2025 4:42:01 PM	10	10	1,000	870	974	-	-	Great service and friendly staff
[Redacted]	[Redacted]	2022	[Redacted]	1/29/2025	2/2/2025 4:41:36 PM	6	8	413	225	375	-	-	The car was in the shop for over a month and the communication was terrible.
[Redacted]	[Redacted]	2020	[Redacted]	1/23/2025	2/2/2025 4:41:15 PM	7	10	977	660	914	-	-	-
[Redacted]	[Redacted]	2024	[Redacted]	1/21/2025	2/2/2025 4:39:55 PM	10	9	985	905	969	-	-	Great staff who will work with you and resolve all your problems.
[Redacted]	[Redacted]	2023	[Redacted]	1/27/2025	2/2/2025 4:36:23 PM	10	10	1,000	985	997	-	-	-
[Redacted]	[Redacted]	2024	[Redacted]	1/30/2025	2/2/2025 4:36:01 PM	10	10	1,000	835	967	-	-	-
[Redacted]	[Redacted]	2024	[Redacted]	1/24/2025	2/2/2025 4:34:35 PM	10	9	976	760	933	-	-	Just good service and everyone was so nice
[Redacted]	[Redacted]	2022	[Redacted]	1/30/2025	2/2/2025 4:34:13 PM	10	10	983	1,000	986	-	-	Mark from the service department has been helping me for multiple years and I trust him and the service team. They always take good care of

Showing 1 to 100 of 433,282 entries

/ boostCX Dealer Dashboard

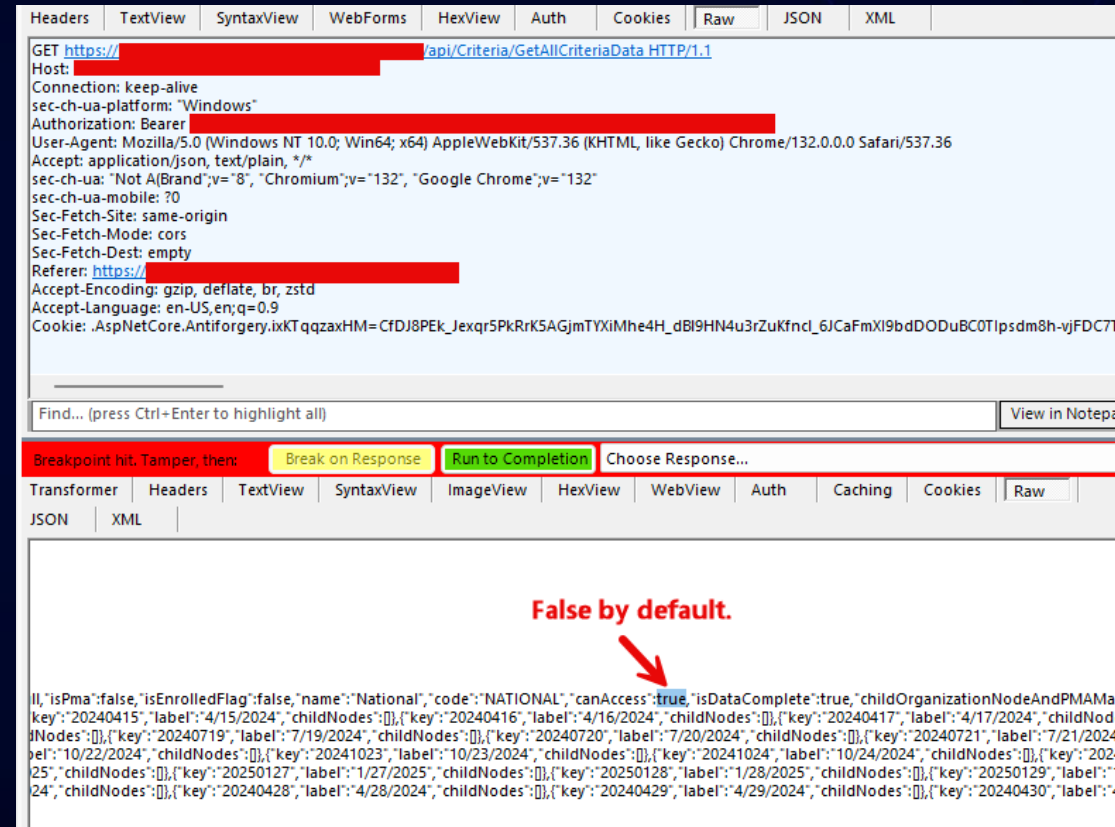


/ Aftersales map for growth

- Solution developed for dealers to identify growth opportunities
- Displays est. location of customers
- National view not functional, hack needed...

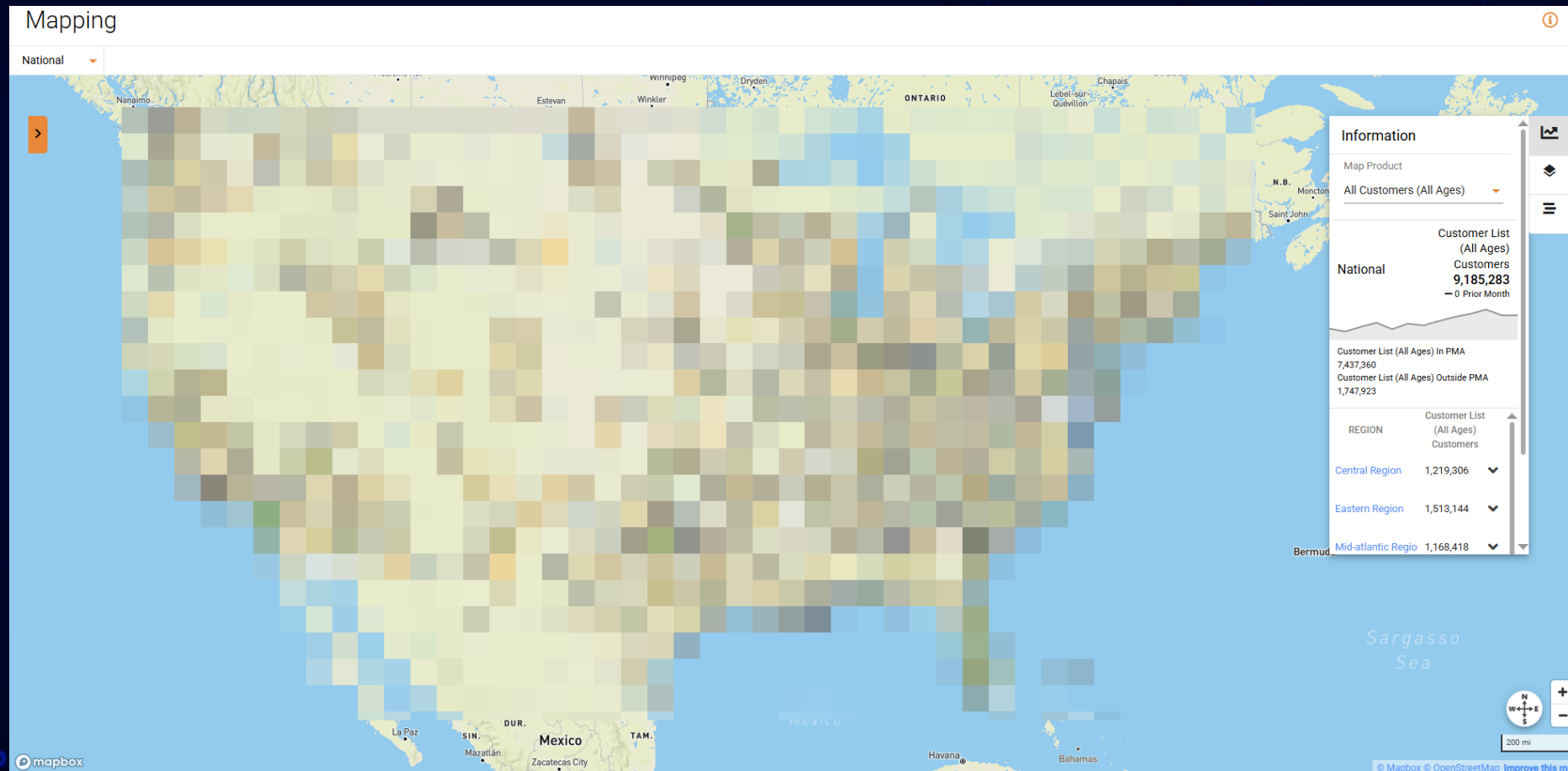
/ Aftersales map for growth – patch

- Fiddler used to intercept, modify HTTP response



/ Behold: the map

- Over 9 million people! So many growth opportunities!



/ Phishing on the dealer's dime



- Marketing platform exists for mail & email
- Ability to create campaign using dealer customer base, bill them for it

Program Information **Campaign Manager** **Reporting**

1 Create Campaign 2 **Select List** 3 Select Content 4 Select Dates 5 View/Approve

Create New Campaign ()

List Attribute Selection

Select the customer segments that will define the desired audience of your campaign.

Customer Segment:

Owner ☒

List Summary **Seed List** **Get Counts**

Media	Customers	Cost/Contact	Est.Cost
Email-null		\$	\$
Email-Owner		\$	\$
Email-		\$	\$
Campaign Setup Fee			\$
Total Customers Count:			
Approximate Production Costs:	\$		

Need Help Reducing Costs? [Get Original Count](#)

Enter # of customers, % of customers or total cost to help meet budget needs.

None **Update List**

< Back **Next >**

Quit Without Saving **Save & Quit**

Program Information **Campaign Manager** **Reporting**

1 Create Campaign 2 Select List 3 **Select Content** 4 Select Dates 5 View/Approve

Update Campaign () Created By: | Created Date: 2/9/2025 2:41:18 PM | Last Updated Date: 2/9/2025 2:41:42 PM

Email Content **Preview PDF**

Subject: Open to see the latest offers.

Preview Text: These exclusive offers are expiring soon so explore your options now!

View Online

Take advantage of these limited time offers now.

2.99% APR or **\$1,000**

for up to 48 months! Dealer Choice Bonus Cash!

/ Disclosure Timeline

- **Jan 26, 2025:** Discovered. Spent about a week exploring, undetected. Very careful to limit actions to read-only.
- **Feb 3, 2025:** Reported through VDP.
- **Feb 6, 2025:** Response received confirming they are investigating.
- **Feb 11, 2025:** Response received confirming fixes, verified.
- **Feb 12, 2025 onward:** Continued collaboration & meetings with vendor to answer questions and coordinate this disclosure.
- **DEF CON 33:** Public disclosure / world premiere.

/ Final Takeaways

- Dealer systems powerful, loaded with your personal info
- Dealer systems more interconnected than ever. Rogue employee = critical impacts
- Simple flaws can have devastating effects on them and YOU
- Even more potential, but had to stop & report eventually
- That's all! Now go rip out your headunits 🛠️



[linkedin.com/in/eatonz/](https://www.linkedin.com/in/eatonz/)



eaton-works.com



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harness

77/77